



Experimental investigation of the fluid-structure interaction during the water impact of thin aluminium plates at high horizontal speed



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ABSTRACT

The water impact of an inclined flat plate and at high horizontal velocity is experimentally investigated with focus on the fluid-structure interaction aspects. Several test conditions have been examined by varying the vertical to horizontal velocity ratio, the pitch angle and the plate thickness. Measurements are performed in terms of strains, loads and local pressure. The study highlights the significant changes in the strains and, more in general, in the structural behaviour when varying the plate stiffness and the test conditions. For some of the test presented, permanent deformations are also found. The strong fluid-structure interaction is analysed by comparing the simultaneous measurements of strains and pressures, and it is shown that the deformation of the plate leads to a reduction of the pressure peak and to a corresponding pressure rise behind it. The variation in the shape of the spray root caused by the structural deformation are discussed based on both pressure measurements and underwater images. Despite the reduction of the pressure peak intensity, it is shown that the structural deformation leads to an increase in the total loading up to 50% for the test conditions examined in this study. It is also observed that in presence of large structural deformations the hydrodynamic loads do not obey the scaling that works in the case of the thick plates, and some practical conclusions about the scaling of tests in presence of a strong fluid-structure interaction are provided.

1. Introduction

In this paper the fluid-structure interaction effects taking place during the water entry of inclined flat thin aluminium plates at high horizontal velocity are investigated experimentally. The basic motivation for the study stems from the aircraft ditching problem, i.e. the emergency landing on water of aircraft in controlled conditions. Although being, fortunately, a rare event, this problem still imposes aircraft manufacturers to carefully consider its occurrence during the design stage, as the aircraft has to be certified at ditching according to the regulations of the airworthiness authorities. Full scale tests on aircraft are of course impractical, being too expensive, therefore scaled model tests are generally considered [1,2]. However, as discussed in [3], scaled model experiments are not fully representative. Generally, the structure is not scaled, thus the main struts and the skin are much stiffer than in reality. The poor reliability of the scaled model tests is also associated with the low test speed, which prevents the occurrence of some important hydrodynamic phenomena [1,4].

Detailed reviews of the studies on water entry problems with application to aerospace structures can be found in [5] and in [6]. In [7] and [8] some problems related to helicopter ditching are examined. The

use of computational approaches for the design and certification of aircraft at ditching is of course very attractive, and significant steps forward have been taken recently [9–13]. Nonetheless, a further enhancement of the capabilities of those tools to reproduce both the hydrodynamics and the strong fluid-structure interaction during the ditching phase is advisable. For such a development, experimental data truly representative of what happens in a real scenario are needed.

Water impact problems are far more common in the naval field, as the ship often moves in a wavy sea (hull slamming), see [14–18]. The impact on the water surface is always accompanied by an intense pressure peak moving along the hull. Nevertheless, in [19] it is shown that the pressure peak on its own, especially when it is large, is quite irrelevant for the determination of the structural response, because it is strongly sensitive to the external conditions and, moreover, is too concentrated in space and time to affect significantly the structural behaviour. In this respect, it is generally observed that the hydro-elastic effects are of great importance, and therefore it is also essential to gather information on the local deformations of the body, using for example strain measurements, and to examine their coupling with the overall velocity and the pressure fields, as well as with the shape of the free surface during the water entry. For some specific cases, it is also

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essential to account for the occurrence of ventilation and cavitation [4,14,20].

The relevance of fluid-structure interaction phenomena during the water entry of a plate or wedge, with focus on slamming, is assessed in several works. In [21] a simplified model to investigate the effect of curvature, characteristic of the bow part of racing yachts, is developed. The hydroelastic interaction during panel-water impact is also studied in [22]. By using numerical simulations, kinematic effects related to the structural deformation of the fluid-solid boundary, and inertia effects, related to the rate at which the structure responds to the external loading, are distinguished. These effects are highlighted by comparing the results with numerical simulations in which the structural problem and the hydrodynamic one are not coupled. In [23] a series of experiments aimed at validating the numerical simulations performed in [22] is presented. It is shown that the largest hydroelastic effect appears to be a time-lag of the peak found in the measured strains with respect to the one derived from the simulations. This time-lag is assumed to be inertia related. Kinematic hydroelastic effects, instead, are mainly concentrated in the upper part of the panel, where a local decrease in the deadrise angle is observed. In [24,25] a comparison between experiments on highly deformable wedges entering water and numerical simulations using SPH (Smooth Particle Hydrodynamics) is presented. It is observed that the intensity of the hydroelastic effects is governed by the ratio of the wetting time to the natural period of the structure, as already pointed out in [14,19]. It is also found that more than one shape mode can be excited. However, at least for the conditions examined, the structural deformation does not appear to influence the hydrodynamic pressure distribution, unless separation occurs as a consequence of cavitation and ventilation, which are phenomena that cannot be captured using their numerical model. Khabakhpasheva and Korobkin [26] developed a semi-analytical method to solve the two-dimensional problem of the water impact of an elastic wedge. Wagner theory is used for the determination of the hydrodynamic loads, and the structural response is derived using modal analysis. Shams and Porfiri [27] developed a novel method for the simulation of the hydroelastic impact of flexible wedges in free fall with arbitrary boundary conditions. The model is able to capture the footprints of the hydroelastic coupling, including the oscillations of the impact forces due to the appearance of the instantaneous suction on the wedge and the pressure fluctuations in the flow field. A rather detailed investigation of the slamming response of stiffened aluminium plates and of the effect of small imperfections generated by the welding process is provided in [28], where some design recommendations are given as well.

All the works cited above refer to the pure vertical water entry case. However, the presence of a horizontal velocity component may affect both the intensity of the pressure and the wetting time, and its effect may be very significant, particularly if the horizontal component is very high, which is the case of aircraft ditching. A theoretical model of the fluid-structure interaction during the water impact of a two-dimensional elastic plate at high horizontal velocity is proposed in [29]. The plate is modelled as an Euler beam with free-free boundary conditions. It is found that the plate deformation may lead to a significant increase in the hydrodynamic loading, and thus the hydrodynamic and structural problems have to be fully coupled in order to achieve a physically accurate result. Wang et al. [30] performed a numerical and theoretical study of the water impact of a horizontal plate with both horizontal and vertical speed. The plate is modelled as a series of longitudinal strips, each one being treated as an Euler beam, and the study is mainly focused on the longitudinal compressive force. It was found that the structural response of the plate and the compressive force strongly depend on the boundary conditions and on the horizontal speed. Experimentally, the water impact with horizontal velocity of a flat plate is investigated in [31], but no information about the structural deformation is available. However, the pressure data measured by the different probes seem well correlated, and therefore it is expected that the plate used for the tests is thick enough to remain in the elastic regime. Water

impact tests at high horizontal velocity of elastic plates are presented in [32] and [33]. The latter work is focused on hydrodynamic aspects, whereas in the former one some aspects concerning the plate deformation are also addressed. Results of tests performed with a horizontal velocity in the range 30 to 45 m/s, vertical velocity of 1.5 m/s and pitch angles from 4 to 10 degrees are provided. In order to assess the test-to-test repeatability, a plate made of aluminium alloy AL2024-T3, 15 mm thick is employed, thus ensuring that the deformations are always within the elastic range and are small enough not to affect the hydrodynamics. Beside thick plates, much thinner aluminium plates, 3 mm and 0.8 mm thick, are also tested. A preliminary analysis of the strains in some test conditions is presented in [34], where significant fluid-structure interaction aspects are highlighted.

In this paper, the structural deformations of the aluminium plates and the subsequent changes in the hydrodynamics are analysed based on the data of the tests performed at different impact velocities, pitch angles and for a different plate thickness. Results are mainly presented in terms of strains. However, for a few cases the simultaneous measurements of both strains and pressures are provided in order to evaluate the changes in the hydrodynamics induced by the structural deformation. The effects of the structural deformation on the hydrodynamics are highlighted by underwater images, whereas more relevant consequences can be observed based on the loads measured in different conditions.

2. Experimental setup and test conditions

Experiments were performed on the High Speed Ditching Facility (HSDF), available at the CNR-INM. The facility consists of a guide, which is suspended on a water basin through five bridges. The guide can be rotated in order to achieve different horizontal to vertical velocity ratios. Detailed information on the facility can be found in Iafrati et al. [32]. The flat plate (the test model) was installed onto a box structure, which was mounted underneath a trolley running along the guide.

It is useful to distinguish the two reference frames used in the following and shown in Fig. 1: an earth-fixed reference frame, with coordinate system (X, Y, Z) , and a reference frame with coordinates (x, y, z) fixed to the plate and moving with the trolley. The horizontal and vertical velocity of the trolley, denoted as U and V respectively, are referred to the earth-reference frame.

A technical drawing of the trolley is shown in Fig. 1. α is the pitch angle, β is the angle of inclination of the guide, which corresponds to $\arctan(V/U)$. The trolley was accelerated to the test speed by a system of elastic ropes, and it was left free to run along the guide a short time before the impact. Tests were performed on aluminium plates of different thickness. Firstly, a plate 15 mm thick was tested: in this case the structural deformations remained within the elastic range, thus allowing to explore a wide range of conditions, assuring a negligible effect of the structural deformation on the hydrodynamics. The results of these tests are detailed in Iafrati et al. [32] and in Iafrati [33]. Such results are referred to as thick plate in the following. In addition, thin panels, 3 mm and 0.8 mm thick, were considered, for which larger and even permanent deformations were expected.

All plates were made of the aluminium alloy AL2024-T3, characterized by a yielding stress of approximately 340 MPa, which corresponds to a yielding strain of about 4600 $\mu\text{m}/\text{m}$. The plates were 500 mm wide and 1000 mm long and were connected to a thicker aluminium frame by two rows of bolts, as shown in Fig. 3, mimicking a quasi-clamped boundary condition all around. The width of the frames was 75 mm and 50 mm for the thick and thin plates respectively. The different width was motivated by the different scopes of the tests. For the thick plates the wider frame was chosen to maximize the stiffness. In the thin plates the frame border was smaller in order to maximize the deformation, still keeping the two-row bolt connection and mimicking the clamped boundary condition as much as possible. Bolts diameters of

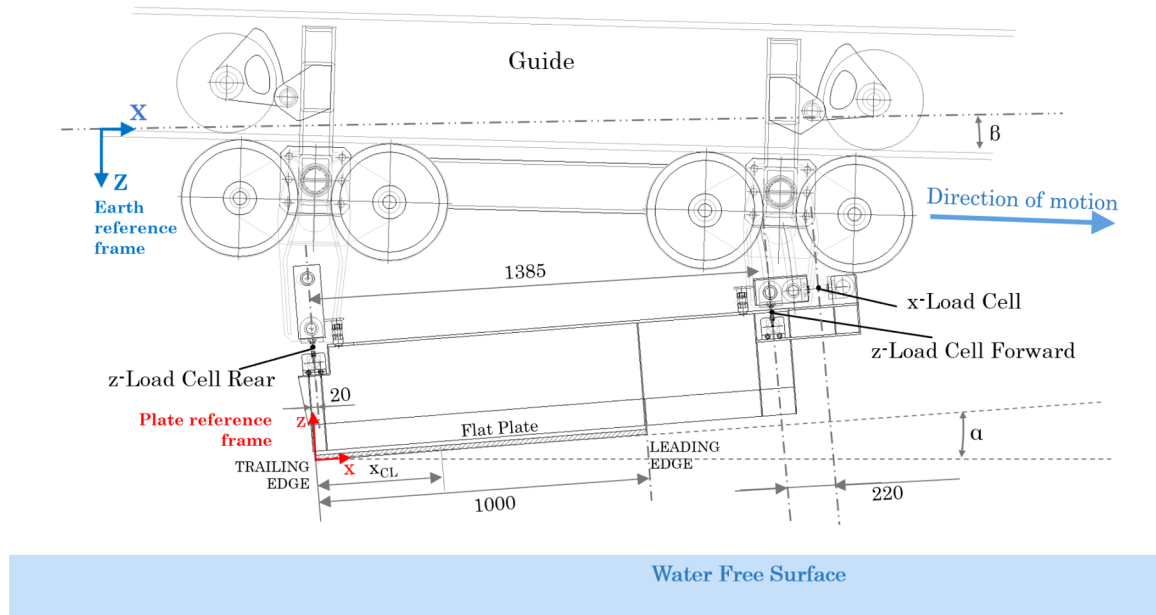


Fig. 1. Technical drawing of the trolley installation on the guide (dimensions are in mm).

5/16" and 1/4" were used for the thick and thin plates, respectively. In order to prevent the pull-out failure at the bolts heads, for the 0.8 mm plates a 3 mm thick reinforcement, 54 mm wide all around, was bonded with a structural adhesive to the plates before connecting them to the frame.

The plates were instrumented by biaxial strain gauges, type Vishay C2A-12-125LT-350 for the thick plates and Vishay EP-08-125TQ-350 on the thin plates, the former allowing a maximum deformation of 3%, the latter arriving up to 15%. Six and eight strain gauges were used for the thick and for the thin plates respectively. In the following S_x and S_y denote the strain measurement in the x and y directions, respectively. In all tests with the thick plates and in some of the tests with the thin plates, pressures were measured using Kulite XTL-123B probes, 300 psi full-scale range. The position of the strain gauges and of the pressure probes is shown in Fig. 2. A picture of a 0.8 mm plate instrumented with the strain gauges only is shown in Fig. 3. The total loads acting on the plates were measured by six load cells: four Kistler piezo-electric load

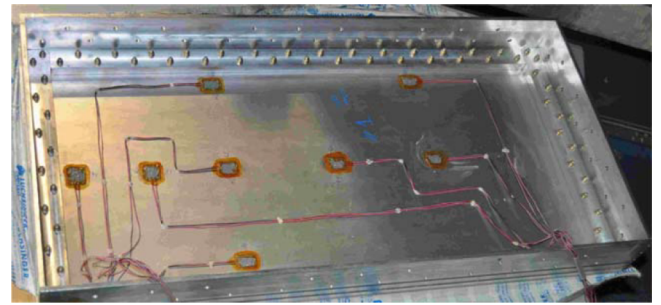


Fig. 3. Flat plate instrumented with the strain gauges. The trailing edge of the plate is on the left.

cells, type 9343A, full-scale range of 70 kN, were used to measure the load component acting normal to the plate (z-direction), whereas two Kistler 9363A, full-scale range of 120 kN, were used to measure the

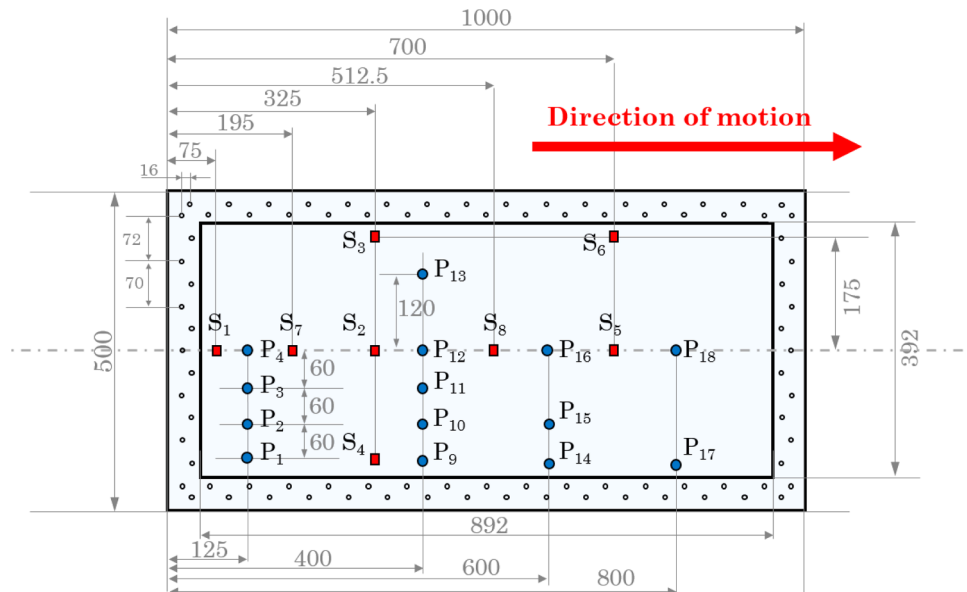


Fig. 2. Top view of the plate with the position of the strain gauges and of the pressure probes. The trailing edge of the plate is on the left (dimensions are in mm).

tangential component (x-direction). The forces are referred to the reference frame fixed to the plate. All signals were acquired by four Sirius and one Dewe43 Dewesoft DAQ modules, connected to a ruggedized computer. Pressure signals were sampled at 200 kHz, whereas all other channels were sampled at 20 kHz. The DAQ modules include an anti-aliasing filter and an A/D converter. After the data have been stored, no further filtering was performed. All measurement are zeroed by taking the average over a short time interval before the impact, and therefore a null value of the pressure corresponds to the atmospheric value.

The tests were recorded by two high-speed cameras, 1024x1024 pixel resolution. One camera, operated at 5000 frame per second (fps), was located at the side and was used to measure the speed at the impact, whereas the other camera, operated at 2000 fps, was located underwater to visualize the impact phase and the flow features beneath the plate.

The tests were performed by varying the horizontal velocity, the vertical to horizontal velocity ratio V/U and the pitch angle. Here attention is mainly focused on the tests performed on the 3 mm and 0.8 mm aluminium plates, but a comparison with the corresponding tests on the 15 mm aluminium plates [33] is also presented in order to highlight the role played by the structural deformation on the loading and on the hydrodynamics. Each test condition is identified by a code **TT_PP_VV_I**. **TT** is a one or two-digit number that indicates the plate thickness (15, 3 and 08 stand for 15 mm, 3 mm and 0.8 mm respectively). **PP** is a two-digit number that indicates the pitch angle at impact (04 for 4°, 06 is for 6° and 10 is for 10°). **VV** is a two-digit number that indicates the nominal test velocity (30 for 30 m/s, 40 for 40 m/s and 45 for 45 m/s). The final letter **I** denotes the repeat in alphabetical order (a, b, ... for the first, second repeat and so on). The complete list of the tests performed on the thick (15 mm) and thin plates (3 mm and 0.8 mm thick) is given in Table 1.

The use of elastic ropes to accelerate the trolley does not allow a precise control of the nominal test velocity, which can be matched with an accuracy of ± 0.50 m/s. A detailed analysis of the test repeatability,

Table 1
Test conditions for the thick (15 mm) and thin (3 mm and 0.8 mm) plates.

Thick Plates					
ID code	Thickness [mm]	U [m/s]	V [m/s]	Pitch Angle [°]	Pressure probes
15_04_45_a	15	44.89	1.5	4	yes
15_04_45_b	15	45.27	1.5	4	yes
15_04_45_c	15	45.27	1.5	4	yes
15_10_30_a	15	29.88	1.5	10	yes
15_10_30_b	15	30.92	1.5	10	yes
15_10_30_c	15	30.74	1.5	10	yes
15_10_45_a	15	45.85	1.5	10	yes
15_10_45_b	15	45.85	1.5	10	yes
15_10_45_c	15	45.85	1.5	10	yes
15_06_40_a	15	40.43	1.5	6	yes
15_06_40_b	15	40.10	1.5	6	yes
15_06_40_c	15	39.78	1.5	6	yes
Thin Plates					
ID code	Thickness [mm]	U [m/s]	V [m/s]	Pitch Angle [°]	Pressure probes
3_06_40_a	3	40.08	1.5	6	no
08_06_40_a	0.8	40.08	1.5	6	no
08_06_40_b	0.8	40.08	1.5	6	no
3_04_45_a	3	46.04	1.5	4	no
08_04_45_a	0.8	45.65	1.5	4	no
3_04_45_b	3	46.04	1.5	4	yes
08_10_45_a	0.8	45.85	1.5	10	no
3_10_30_a	3	30.92	1.5	10	yes
08_10_30_a	0.8	31.10	1.5	10	no
3_10_30_b	3	30.92	1.5	10	yes
08_10_30_b	0.8	31.28	1.5	10	yes

based on the results obtained for the thick plates, is provided in [32]. Therein, the data of ten repeats of the tests performed at two different pitch angles were used to retrieve the ensemble average and the standard deviation in terms of pressures, strains and loads. A similar analysis was performed on the thin plates, even though, due to the occurrence of permanent deformations, the number of repeats was limited to two. More details on the test repeatability are found in Appendix B.

In order to compare the results of the different tests or repeats, an origin of the time axis has to be chosen. For the thick plates the origin of the time was set at the passage of the pressure peak at the pressure probe P_4 , i.e. the first probe located along the midline. For the thin plates, since pressures are only available for a limited number of tests, the peak of the strain S_{1x} is used instead. Of course, the first contact of the plate with the water surface happens somewhat before the chosen time origin, as it can be argued from the growth of the normal load measured at the rear. It is worth remarking that when comparing the results of thick and thin plates, a small misalignment in the time histories may occur due to the different positions of the strain gauge S_1 . For the purpose of the study, the most interesting part of the test is the so-called 'impact phase', which lasts from the first contact of the plate with water up to the time at which the spray root reaches the leading edge, and the front face of the acquisition box impacts with water, thus leading to a sharp deceleration. In the following, the beginning of the impact phase is assumed to coincide with the time origin defined above, whereas the end is identified from the test data as the time t_E , at which a sharp rise in the tangential load F_x occurs.

3. Experimental results and discussion

3.1. Strain measurements

The present study is mainly aimed at investigating how the plate deformation changes when varying the plate stiffness and the test conditions. It is worth noticing that in some conditions permanent deformations occur. As anticipated, for the specific material adopted in the tests, the yielding limit is estimated to be about $4600 \mu\text{m/m}$. It is expected that fluid-structure interaction phenomena can have a significant effect on the dynamics of the impact for the thin plates (3 mm and 0.8 mm). This is due to both the low thickness and to the low value, less than or equal to 1, of the ratio of the wetting time to the first natural frequency of the plate, which is estimated in Appendix A.

As already mentioned, the attention is focused here on the impact phase only. However, the plate is still highly loaded at the end of the impact phase, which is the reason why the strains do not return to zero even for the thick plates. This is clearly seen in Fig. 4, where the time histories of S_{5x} for the cases XX_06_40, i.e. $U = 40$ m/s, $\alpha = 6^\circ$ and different thickness (XX = 15, 3 or 08), are drawn for a time interval much longer than the impact phase. The data indicate that in all cases the strain is non-zero at $t = t_E$, whereas it returns to zero for $t \gg t_E$ for the 15 mm and 3 mm plates only. For the 0.8 mm plate, the yielding limit is exceeded, and the strain remains at a finite non-zero value at the end of the tests. It is worth noticing that out-of-plane deformations are also observed for the 3 mm plate, in some cases at least. However, they were much smaller than those observed for the 0.8 mm plate, and they were mainly localized close to the frame, which explains why no residual strains appear in the middle of the plate.

In Fig. 5 the longitudinal strain measured by the gauges located along the midline are shown both for the thick and thin plates, for the test conditions XX_10_30, XX_06_40 and XX_04_45 (i.e. for XX = 15, 3, 08). In order to help the reader, a drawing of the gauge installation is provided in Fig. 5(a) with the position of the gauges indicated with the same colour used in the plots. For the thin plates, i.e. the 3 mm and 0.8 mm plates, in order to facilitate the data comparison among the different test conditions, the same vertical scale is adopted. The same vertical range, albeit much smaller than that used for the thin plates, is also adopted for the comparison with the thick plate data. As anticipated in Section 2, the

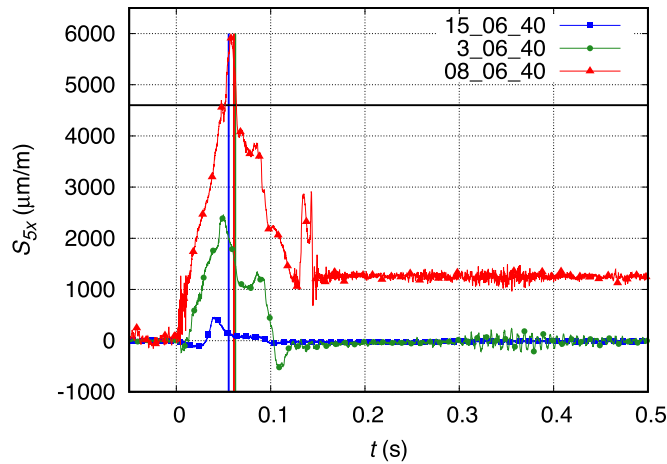


Fig. 4. Time histories of S_{5x} for the cases XX_06_40, i.e. at $U=40$ m/s, pitch angle 6° and different thickness (i.e. XX=15,3 or 08). The end of the impact phases for each test is indicated by vertical lines of the same colour as that of the corresponding plot line. The black horizontal line indicates the value of the yielding strain.

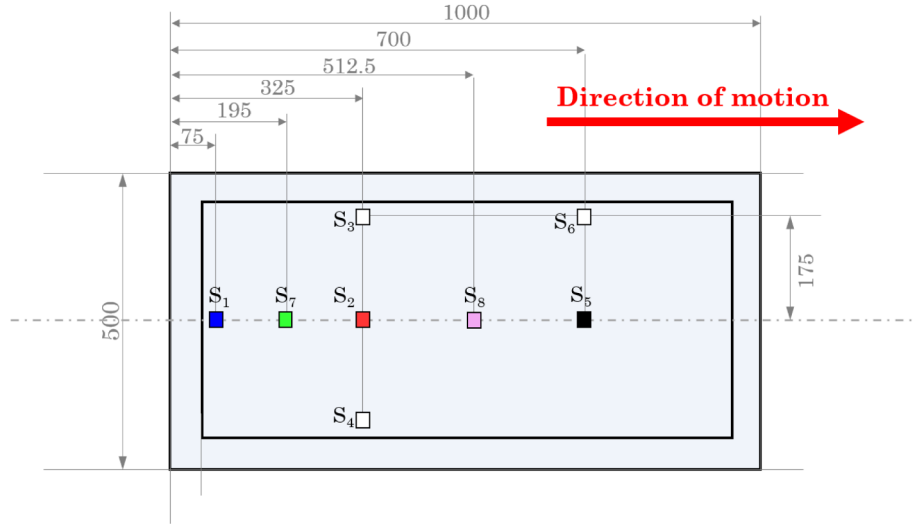
origin of the time axis is taken at the first peak of strain gauge S_{1x} . Strains are drawn versus t/t_E with t_E denoting the end of the impact phase for each condition, so that all data are meaningful in the same range $t/t_E \in [0, 1]$. From Fig. 5 it is seen that for the thick plate (15 mm) the strains exhibit a quite similar behaviour in the different cases. For the gauges located in the middle of the plate, i.e. S_2 and S_5 , the x-strains start from zero, diminish in the early stage, then rise up to a maximum and gently decay to zero afterwards. For the gauge S_1 , located very close to the frame and therefore strongly affected by the boundary conditions, the x-strain rises rapidly, then decays gently, reaches a minimum and rises again to zero. It is worth noticing the asymmetry of the strain time histories about the peak. As shown later in Section 3.3, the peak in the strains occurs at about the time when the pressure peak passes beneath the gauge. The pressure is zero ahead of the spray root, rises quite sharply up to the peak and decays more gently afterwards [33]. The lower decay rate, which results in a higher loading of the plate, is the reason for the higher strain levels measured after the passage of the pressure peak. The behaviour is quite different for the thin plates. The strain values are, of course, much higher and, furthermore, at least for the gauges far from the boundaries, the strain level remains constant or diminishes rather slowly after the passage of the pressure peak. Partly, this is due to the fact that the deformation is in the plastic regime, particularly for the 0.8 mm plates, when the yielding limit is exceeded. The flatter strains are also a consequence of the large out-of-plane deformation, which disrupts the spray and causes a spreading of the pressure distribution. Especially for the 0.8 mm plates some hydroelastic effects occur, as it can be seen for $t/t_E \geq 0.4$. The longitudinal strain measured by gauge S_1 is much different from that measured by the other probes along the midline. For the 3 mm plates it shows a peak and decays to negative values afterwards, whereas for the 0.8 mm plates the strain exceeds the yielding limit and remains positive for the remaining part of the test. The strain signals exhibit an oscillatory behaviour, which is sometimes more evident after the first peak. This can be a consequence of the fluid-structure interaction.

The test conditions of the data provided in Fig. 5 do not allow to distinguish the effects of the pitch or of the horizontal velocity. In fact, considering the limited number of tests available for the deformable specimens, the test conditions were chosen in order to be as representative as possible of a realistic scenario. Hence, tests at lower horizontal speed were conducted at a higher pitch angle, which is how the aircraft operates in order to guarantee the same lift. Nonetheless, from Fig. 5 it can be seen that the pitch angle and the test speed do not affect much the strains for the 15 mm and 3 mm plates, whereas they

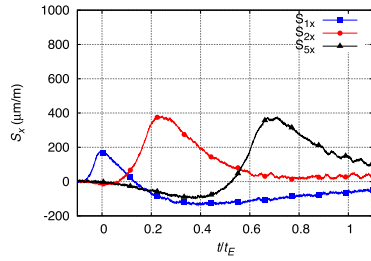
have a quite significant effect on the thinnest plates, particularly for the gauges located in the middle of the plate. At least for the thinnest plates, the effects of the pitch angle and test speed can be distinguished by comparing the data of the test conditions 08_10_30 and 08_04_45 with 08_10_45, as shown in Fig. 6. In particular, the comparison in Fig. 6(a) shows that the longitudinal strains, especially for the gauges S_2 and S_8^1 , increase with the horizontal speed and with the pitch angle. The larger strain is of course associated with a higher loading. As discussed in [33], the loads during the water impact of the thick plates scale with the factor $\rho(U^2 + V^2)\sin(\alpha + \beta)$, so a load increase with U and α could reasonably be expected also for the thin plates. It is interesting to scale the longitudinal strains with the same quantity, as shown in Fig. 7, where the scaled strains are denoted by S_{xs} . In particular, Fig. 7(a) shows that especially S_{2xs} and S_{8xs} at 10° and $U=45$ m/s are higher than those at the same pitch angle and at $U=30$ m/s. Such a behaviour is expected because, as discussed in Section 3.5 and shown in Fig. 24(c), the scaled normal load is higher at 10° and $U=45$ m/s. The interpretation of the effect of the pitch angle is, instead, less straightforward. As observed in Fig. 7(b), the scaled strains S_{xs} are nicely overlapped. As for the other strain time histories, there is an initial phase in which all the scaled strains are overlapped. Successively, there is an intermediate phase, in which the scaled strains at 10° are higher than those at 4° . In the final part of the impact phase, the strains at 4° overcome those at 10° . This is especially evident for S_{2xs} . This behaviour is in agreement with what shown in Fig. 24(c), where it is observed that the scaled load at 4° and $U=45$ m/s results higher than that at 10° and $U=45$ m/s. It is worth noticing that, however, the gap between the scaled strains, especially in the last part of the impact phase, could be increased by the fact that at 10° the velocity reduction within the impact phase (about 2 m/s) is slightly higher than at 4° (about 0.5 m/s), as detailed in [32]. Even though such a velocity variation cannot be easily considered in the scaling, it is estimated that it may be responsible for an increase of about 10% of the scaled strains measured at 10° , which reduces the gap.

In Fig. 8 the y-strains measured by the gauges located along the midline are shown both for the thick and thin plates for the test conditions, i.e. XX_10_30, XX_06_40 and XX_04_45. For the thick plate, as observed already for the strains in the longitudinal direction, the y-strains display a maximum followed by a gentle decay. However, differently from the longitudinal strain, the strains in the transverse direction remain always positive, both before and after the passage of the peak, indicating that the out-of-plane deformation is always upwards. Moreover, the strains are more symmetric about the peak, which is a consequence of the more uniform pressure loading in the y-direction compared to the x-direction. By comparing the above data with the corresponding strains measured in the longitudinal direction, shown in Fig. 5, it is observed that the peak in the y-direction is delayed with respect to the one measured in x-direction, which is of course a consequence of the pressure distribution. The strain level increases as the plate thickness is reduced. For the 3 mm plate, the gauges located far from the boundaries display a gentle rise followed by a peak. After the peak, the strains undergo a small reduction and remain almost constant until the end of the impact phase. The gauge S_1 exhibits a small rise and then diminishes, reaching an almost constant negative level. Much more irregular are the time histories of the strains recorded on the 0.8 mm plates. Gauges S_{1y} and S_{7y} exhibit an initial rise, a peak and a decay to zero. The other gauges show negative strains, quite high in amplitude in the early phase, denoting a downward curvature. Next, the strains rise up to a positive peak and decay to zero. For the 15 mm and 3 mm plates the peak of the strains have about the same amplitude for the different gauges, whereas for the 0.8 mm plate the strain level decreases moving towards the leading edge. The fact that strains tend to

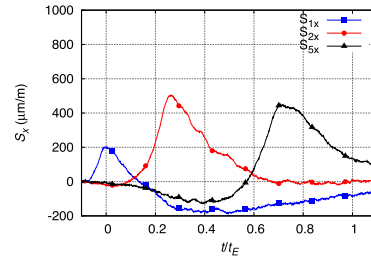
¹ The data recorded by the gauge S_8 for $t \in (0.8, 0.975)$ present a spike, therefore they were not plotted



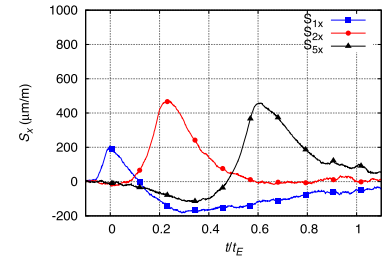
(a) Strain Gauge Positions



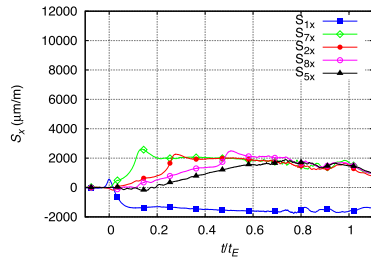
(b) 15_10_30



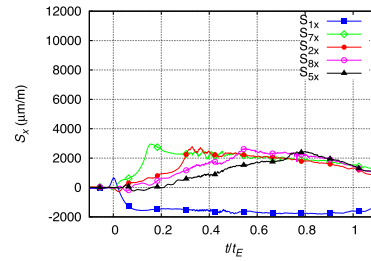
(c) 15_06_40



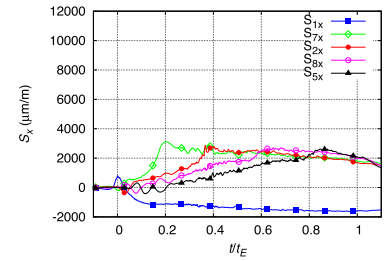
(d) 15_04_45



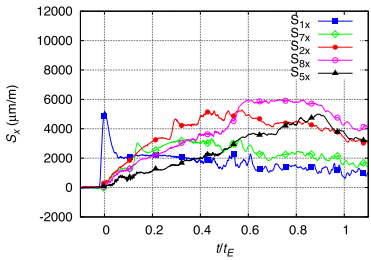
(e) 3_10_30



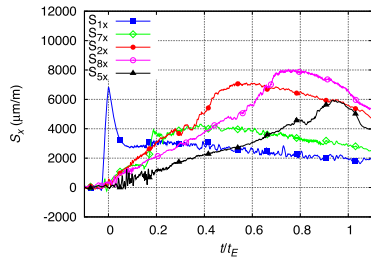
(f) 3_06_40



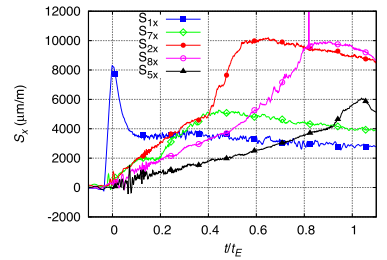
(g) 3_04_45



(h) 08_10_30

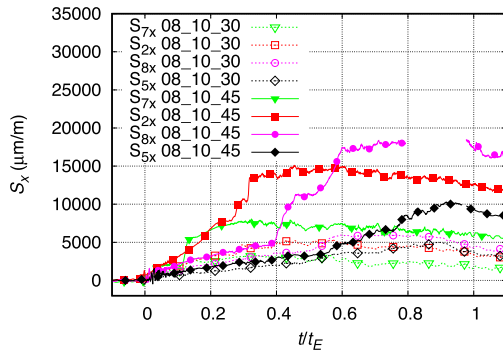


(i) 08_06_40

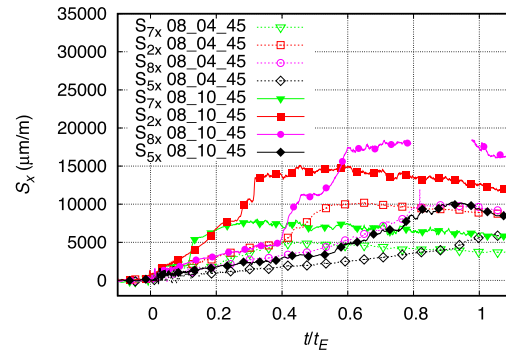


(j) 08_04_45

Fig. 5. Comparison of the longitudinal strains measured along the midline for the test conditions XX_10_30, XX_06_40 and XX_04_45.

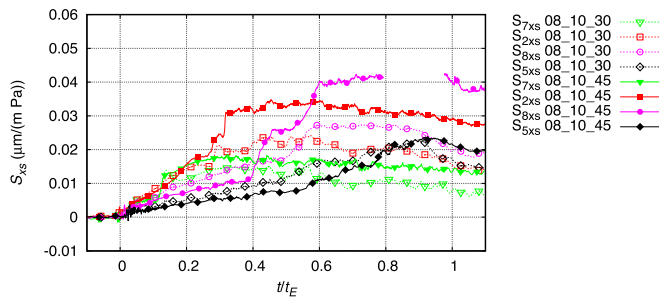


(a) 08_10_XX

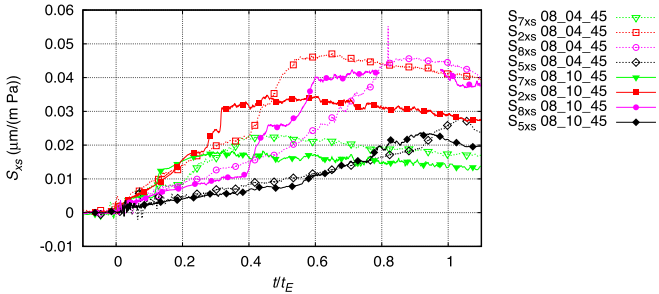


(b) 08_XX_45

Fig. 6. Comparison of the longitudinal strains measured along the midline (a) between the test conditions 08_10_30 (dashed lines) and 08_10_45 (solid lines) and (b) between the test conditions 08_04_45 (dashed lines) and 08_10_45 (solid lines).



(a) 08_10_XX



(b) 08_XX_45

Fig. 7. Comparison of the scaled longitudinal strains along the midline (a) between the test conditions 08_10_30 (dashed lines) and 08_10_45 (solid lines) and (b) between the test conditions 08_04_45 (dashed lines) and 08_10_45 (solid lines).

return zero at the end of the impact phase means that the deformation is basically elastic. The decreasing value of the strain peaks is associated with the reduction of the pressure peaks, which is observed while going towards the leading edge, as discussed in more detail in Section 3.3. At a given thickness, the effects of test speed and pitch angle on the transverse strain, although present, it is not as relevant as for the longitudinal strains, apart from the higher peak of S_7 for the 0.8 mm thickness.

In order to investigate the behaviour of the strains at a fixed distance from the trailing edge, the data of the gauges S_2 , S_3 and S_4 are provided in Fig. 9 and 10 for the x and y components, respectively. As above, the data for the different plate thickness are shown for the three different test conditions. In the x -direction, due to the boundary conditions, the strains measured in the middle are higher than those at the sides. For the 15 mm plate the strain in the middle is about twice as that

at the sides. Something similar also happens for the 0.8 mm plates, at least until the peak is reached at the side. Next, the strain at the side diminishes, whereas the strain in the middle continues to grow for half the duration of the impact phase. This is more evident if the test speed increases and if the pitch angle reduces. A quite different behaviour characterizes the strains for the 3 mm plate. In this case the strain in the middle exhibits a positive growth before the passage of the pressure peak and remains almost constant, or diminishes very slowly, after the pressure peak. On the contrary, the x -strains at the sides show a small peak associated with the passage of the pressure peak, but their values are always rather small and are negative for most of the time. It is worth noticing that the peak of the strains for the thin plates are slightly delayed compared to the 15 mm plate. This is presumably due to the larger out-of-plane deformation of the plates, which induces a change in the pressure distribution, as described in Section 3.3. In all cases, a quite satisfactory overlapping of the measurements at the two sides can be observed, which confirms the symmetry of the loading and, thus, of the test conditions. The data shown in Fig. 9 also confirm that the effect of pitch angle and of the horizontal speed on the longitudinal strains is much more relevant in the middle (S_{2x}) than at the sides (S_{3x} and S_{4x}).

Strains in the y -direction are shown in Fig. 10 for the three test conditions XX_10_30, XX_06_40 and XX_04_45. As already found for the gauges located along the midline, also at these locations the strains in y -direction are smoother than those in the x -direction. For the 15 mm plates, the peak of the strains in the middle of the plate is about three times higher than that at the side, and there are no substantial changes with the test condition. In the 3 mm plates there is a quite a significant difference between the strains in the middle, which are positive, and those at the sides, which are negative, thus denoting a difference in the curvature and the presence of an inflection point. The passage of the pressure peak is not so evident in the curves. Also in this case there are no significant changes in the trends and in the peak values when varying the test condition. The most remarkable difference concerns a delay in the occurrence of the peak at S_2 when increasing the speed and reducing the pitch angle. A completely different behaviour characterizes the y -strains at the side for the 0.8 mm plates. In all three cases the peak values at the sides are comparable to those in the middle. The peak in the middle is delayed with respect to the sides, which is an effect of the forward bending of the spray root, as it is further discussed in Section 3.3. The fact that the strains are rather uniform in the transverse direction is because, due to the small thickness, the effect of the boundary condition is confined to a small neighbourhood of the frame. The data for the 0.8 mm plate also show that the strains, which are negative before the passage of the pressure peak, turn positive afterwards. In a spatial reference frame, this indicates that the strains are positive behind the pressure peak and are negative ahead of the spray root. Hence, for such a small plate thickness, the pressure peak

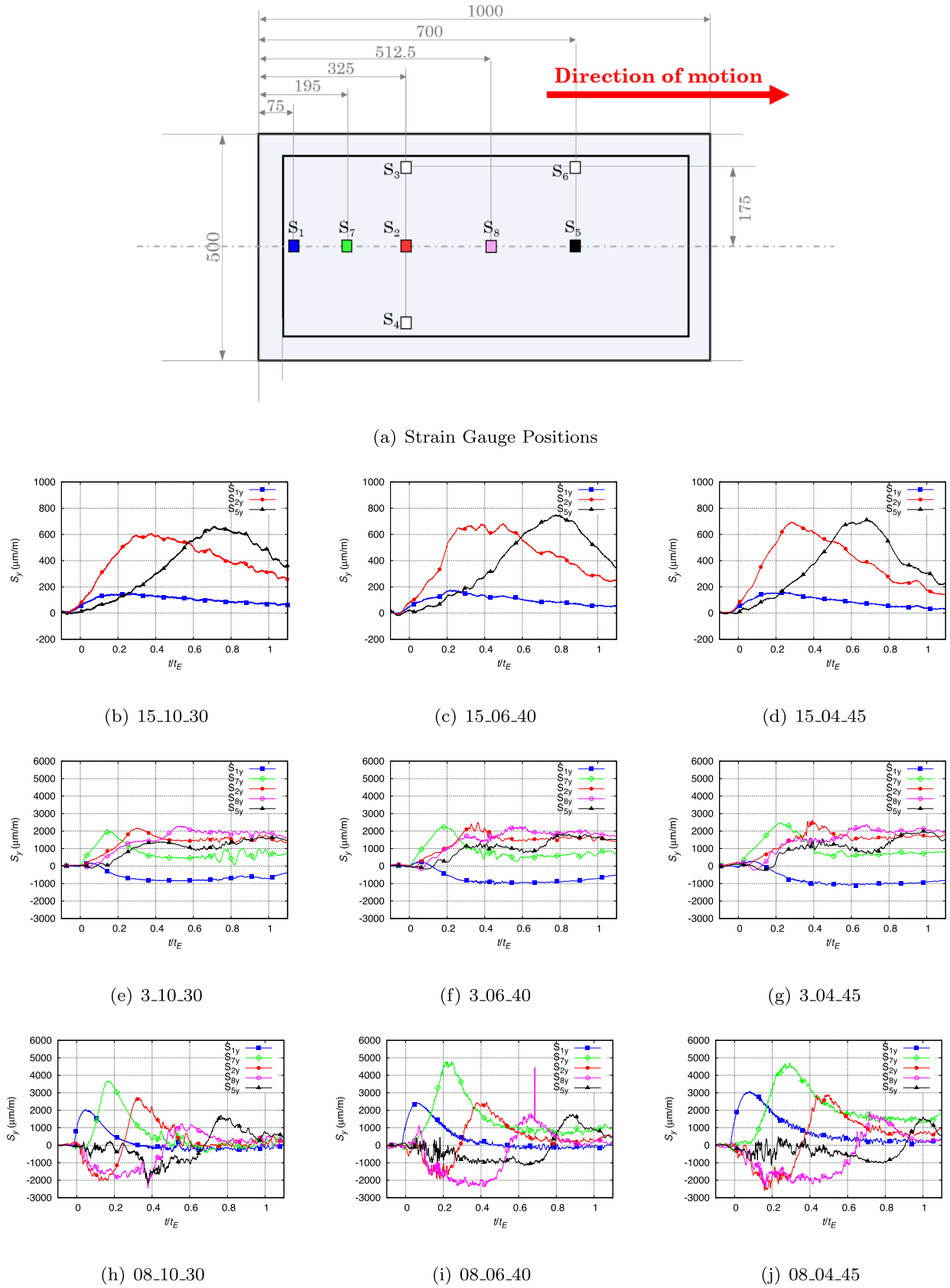
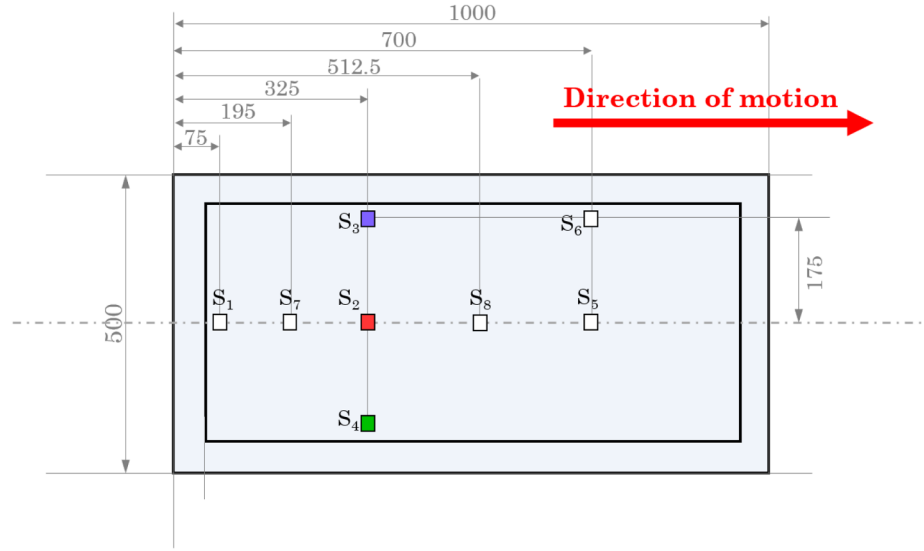
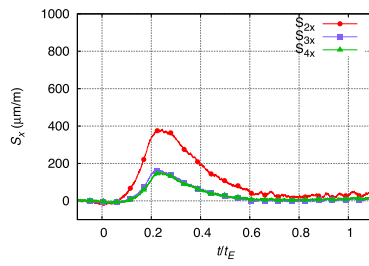


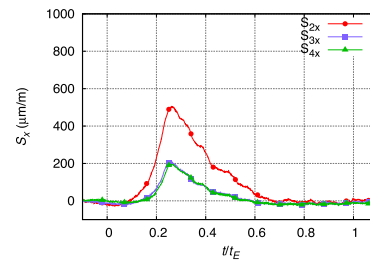
Fig. 8. Comparison of the transverse strains measured along the midline for the test conditions XX_10_30, XX_06_40 and XX_04_45.



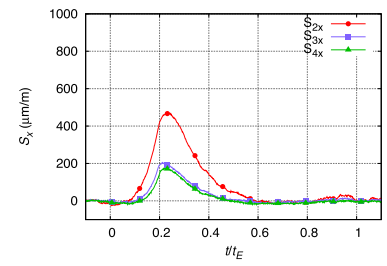
(a) Strain Gauge Positions



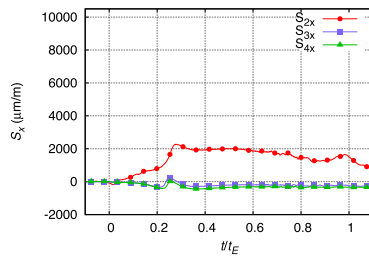
(b) 15_10_30



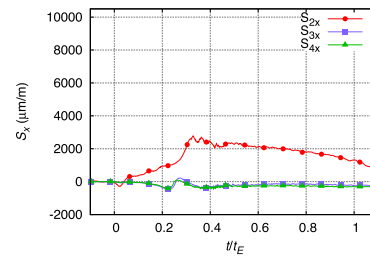
(c) 15_06_40



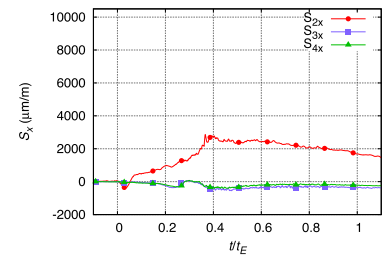
(d) 15_04_45



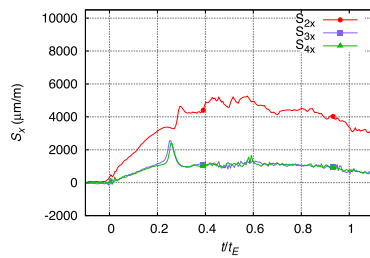
(e) 3_10_30



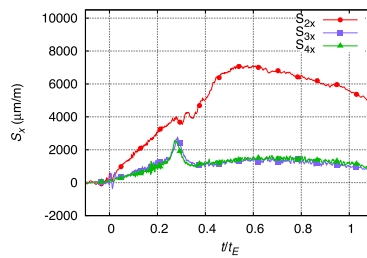
(f) 3_06_40



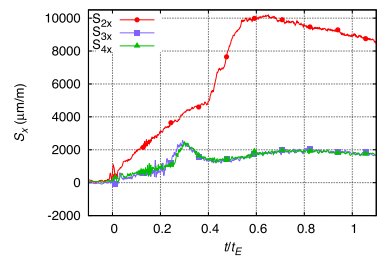
(g) 3_04_45



(h) 08_10_30

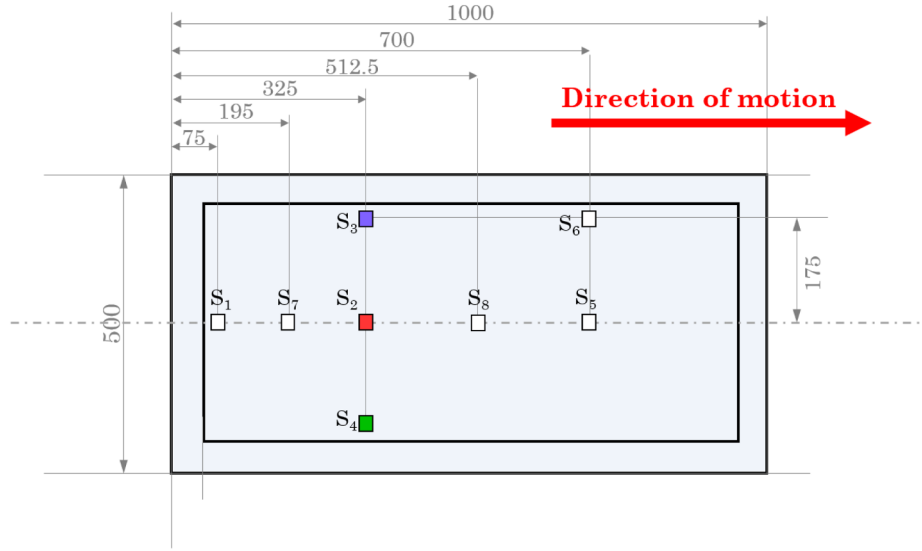


(i) 08_06_40

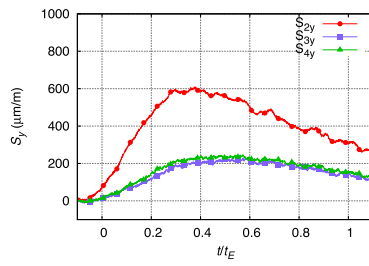


(j) 08_04_45

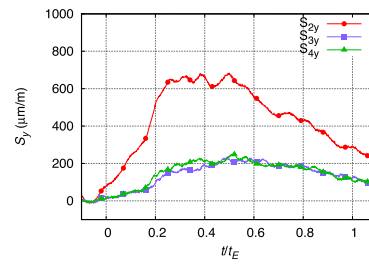
Fig. 9. Comparison of the longitudinal strains measured at $x=0.325$ m for the test conditions XX_10_30, XX_06_40 and XX_04_45.



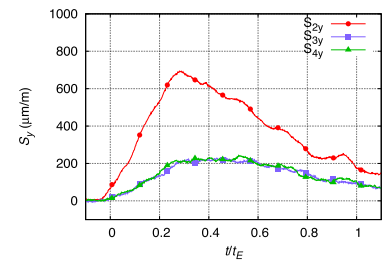
(a) Strain Gauge Positions



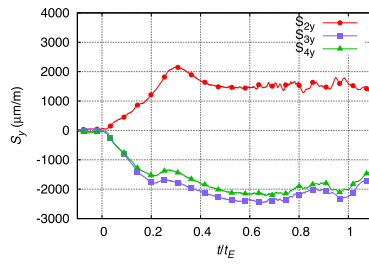
(b) 15_10_30



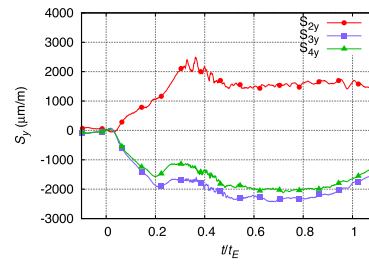
(c) 15_06_40



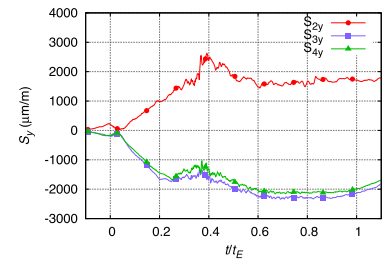
(d) 15_04_45



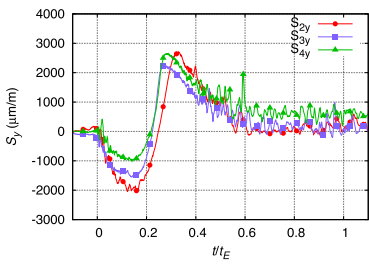
(e) 3_10_30



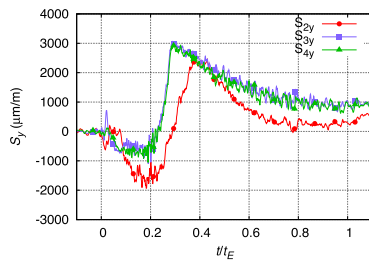
(f) 3_06_40



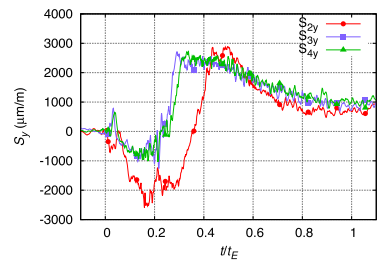
(g) 3_04_45



(h) 08_10_30



(i) 08_06_40



(j) 08_04_45

Fig. 10. Comparison of the transverse strains measured at $x=0.325$ m for the test conditions XX_10_30, XX_06_40 and XX_04_45.

generates an inflection of the plate, which propagates with it.

3.2. Residual deformations

The thin plates display residual deformations at the end of the tests, thus indicating that the strains overcome the yielding limit in some areas. In the previous section it is shown that this is revealed by some of the gauges on the 0.8 mm plates at least. However, residual deformations are also observed for the 3 mm plates, even though the measured strains are always below the yielding limit. Very likely, the maximum strains occur next to the inner boundary of the frame and are not recorded by the installed gauges, because for the thin plates they are located at 25 mm from the boundary itself, see Fig. 2.

The residual deformation is provided in terms of the out-of-plane displacement w measured on a matrix of points upon the plate, before dismounting the plate from the frame. It is worth remarking that the plate is loaded also after the impact phase. Although the highest normal loads are generated during that stage, it cannot be excluded that, for the 0.8 mm plates in particular, the residual deformation is affected by the loading and by the fluid-structure interaction taking place in the final stage of the test.

The colour contours of the out-of-plane displacement are shown in Fig. 11, where the position of the measurement points are indicated with black round circles. Positive values mean upward displacement. The data for the 3 mm and 0.8 mm plates are shown on the left and right columns, respectively. Different scales are adopted for the two cases. In all cases, the displacement is zero in the bolted region, which is 50 mm wide all around. The data show that for the same plate thickness the residual deformations increase if the plate loading increases, hence with the horizontal speed and the pitch angle. The residual deformations also increase if the plate thickness is lower, see also Section 3.1 and Section 3.5.

In some cases the thin plate exhibits a residual bump in the middle, and this is particularly evident in the test 08_10_30_a and to a smaller extent in the test 08_06_40_a. A picture of the plates after these tests is shown in Fig. 12. The formation of the bump can be either the result of buckling or a consequence of the suction forces induced by the vertical oscillations of the last portion of the guide at the end of the test. Partly, the bump is also a consequence of a small shift of the plate beneath the bolts, which is not restored at the end of the test.

The relationship between the strains and the residual deformations can be analysed in more detail using the second derivatives of the out-of-plane displacement in the x and y direction, which are directly related to S_x and S_y , respectively. The signs of the curvature are based on the coordinate system fixed to the trolley and shown in Fig. 1. Using this sign convention, a negative (upward) curvature is associated with a positive value of the measured strain, whereas a positive (downward) curvature is associated with a negative value of the measured strain. The colour contours of the second derivatives for the test conditions 3_10_30, 08_04_45 and 08_10_45 are shown in Fig. 13. The second derivatives are computed from the original measured displacement grid by a double application of a central finite difference scheme, apart from the point at the sides, where a single-sided scheme is used. Of course, due to the poor resolution of the data, the second derivative cannot be very accurate. Looking at the second derivatives in the x direction for the test condition 08_10_30, there is a region with downward (positive) curvature at the trailing and leading edges. This is consistent with the negative value of the time history of S_{1x} , shown in Fig. 5(a). The central area exhibits an upward curvature (negative), which is consistent with the positive values recorded by other strain gauges. The transition from an area of positive to an area of negative second derivative denotes the presence of an inflection point in the permanent deformation. In the case 08_04_45 a similar behaviour of the second derivative is observed. As for the second derivative in the y direction, in both cases 3_10_30 and 08_04_45 the appearance of an area of downward (positive) curvature is quite evident at the top and at the bottom sides. This is in

agreement with the strain time histories shown in Fig. 10(e) for the case 3_10_30, where S_{3y} and S_{4y} display large negative values over the whole impact phase. This could lead to a permanent deformation, even though the recorded strains are not observed to overcome the yielding limit. For the case 08_04_45, shown in Fig. 10(j), however, the time histories of S_{3y} and S_{4y} are always positive, which would imply an upward curvature during the impact phase. The discrepancy could be due to the stress and the loads experienced by the plate after the impact phase, as mentioned above. Finally, the case 08_04_45 shows that the formation of the bump creates significant changes in the plate curvature, as explained above.

3.3. Pressure measurements

As discussed in [33] as soon as the plate touches the water, a thin spray forms and propagates along the plate. A sharp pressure peak is formed just behind the spray root, with the peak intensity being proportional to the square of the spray root propagation velocity [33,35,36]. The aim of this section is to show how and to which extent the structural deformation changes the pressure distribution and the loading. It is worth noticing that pressure probes were used in few selected cases only (see Table 1). This is because before performing the tests, the possibility of a rupture of the plate during the impact could not be excluded, and there was also the risk that the holes needed for the installation of the pressure probes could trigger the rupture itself. As shown below, the results presented in this work indicate that the presence of the sockets and of the probes do not introduce any significant change in the structural response.

3.3.1. Pressure changes in the longitudinal direction

In order to show the changes in the pressure distribution caused by the structural deformations, in Fig. 14 the time histories of pressure measured by the probes located along the midline (left column) and at the side (right column) of the plate are provided for the three different plate thicknesses. The comparison is established for the test condition XX_10_30, i.e. $U=30$ m/s, $\alpha=10^\circ$, which is the one with the lowest loading. For the reader convenience, the position of the pressure probes on the plate is given in the figure on top. Some of the pressure time histories shown in Fig. 14 display a short negative transient just before the sharp peak. This phenomenon is discussed in detail in [32] with the support of underwater videos, and it is found that such negative values are associated with the formation of vortices from the probe heads, which introduce an irregularity in the plate surface. At such a high speed, the pressure in the core of the vortices drops down to the vapour pressure. As a consequence, cavitation above the pressure probe occurs concurrently with the passage of the thin spray. The irregularity in the plate surface is also responsible for the positive pressure just before the sharp rise at some probes, which is presumably due to the formation of a stagnation point, still associated with the same type of vortices [32]. The pressure time histories plotted in Figs. 14(d) and 14(f) show that in the 3 mm plate the peaks at the side, i.e. P_1 , P_9 and P_{14} , are slightly delayed compared to those observed for the thick plate, whereas no substantial changes are found in terms of the peak amplitude. A more significant reduction of the pressure peak is observed instead for P_{17} . The situation is different for the probes located along the midline. In this case the delay is more evident, particularly for P_{12} and P_{16} and, more importantly, the peak is much lower for P_{16} and completely disappears for the probe P_{18} , see Figs. 14(c) and (e). This is of course a consequence of the structural deformation, which is more pronounced in the middle of the plate than at the sides, where the thick aluminium frame and the bolt connection prevent the plate deformation. The deformation of the plate grows as the plate penetrates into the water, and the total load on the plate increases. This is the reason why the changes are more pronounced for P_{16} and P_{18} rather than for P_{12} . The effect of the plate deformation on the pressure measurements is of course more evident in the 0.8 mm plate, as shown in Figs. 14(g) and (h). Even if a

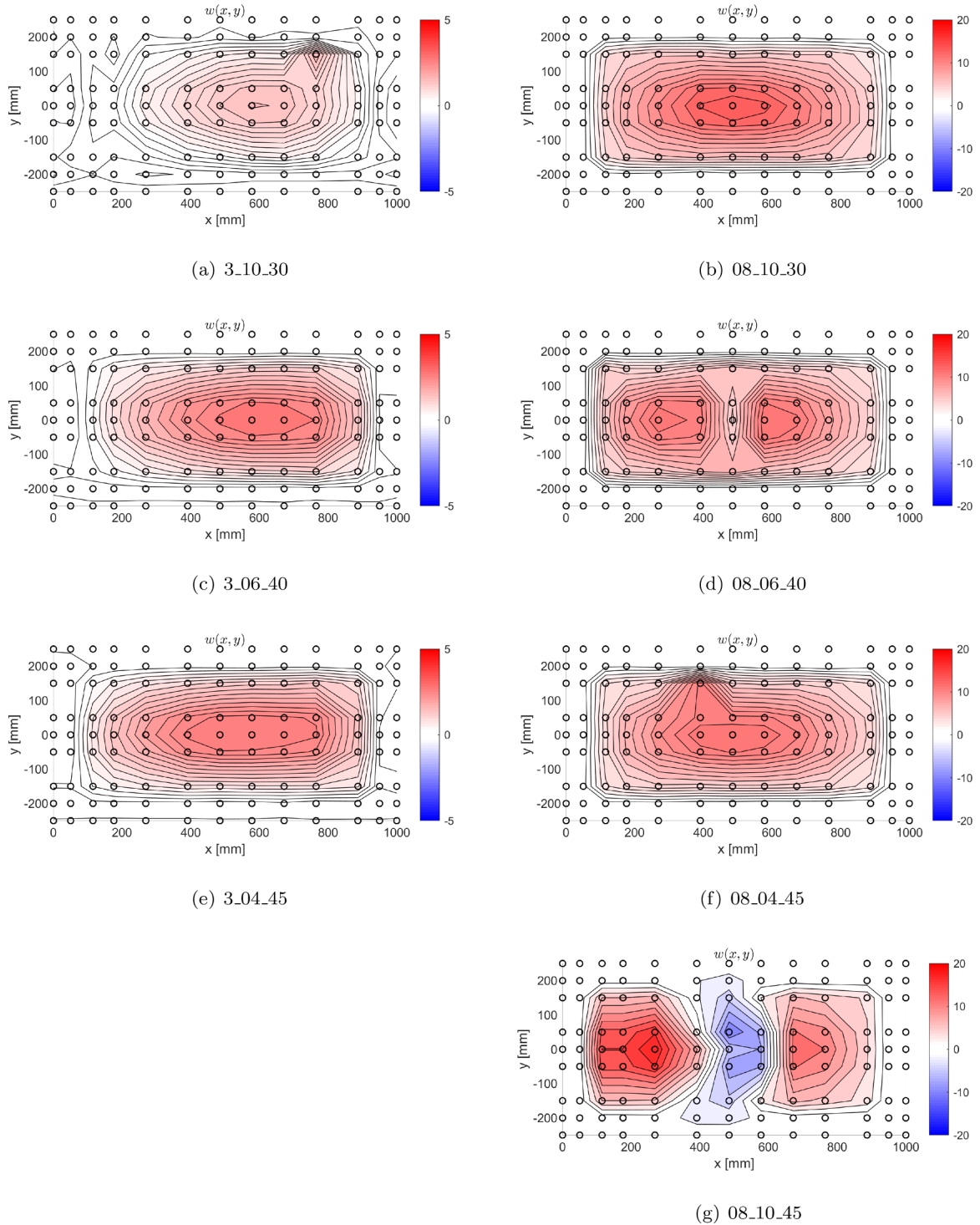
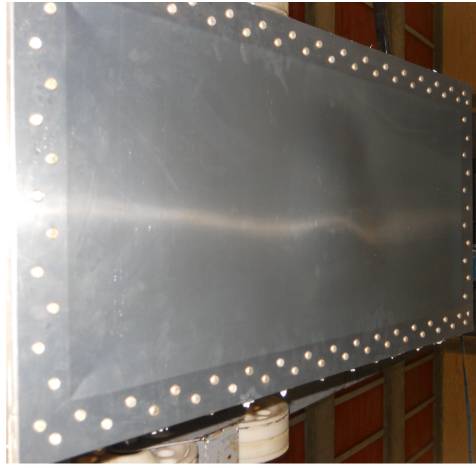


Fig. 11. Residual out-of-plane displacements (mm) measured on the plates at the end of the tests. The trailing edge, corresponding to $x = 0$, is on the left. The black round circles indicate the positions of the measurement points..

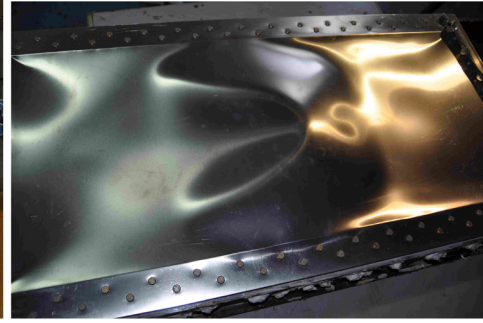
pressure peak, albeit of smaller amplitude, still occurs at the side, e.g. at P_9 for instance, it completely disappears in the middle except for P_4 , which is very close to the boundary. This indicates that the classical spray observed for the thick plates is disrupted by the structural deformation, and the resulting loading becomes more irregular.

Similar comments can be made for the cases XX_04_45, i.e. $U = 45$ m/s, $\alpha = 4^\circ$, see Fig. 15. For this test condition, pressures have been measured for the 15 mm and 3 mm plates only (see Table 1). Owing to the higher velocity of such test condition, the pressure peaks measured for the thick plate are higher than those recorded in the

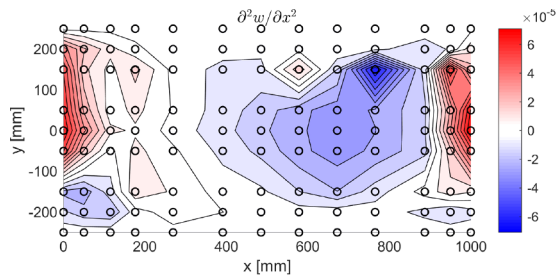
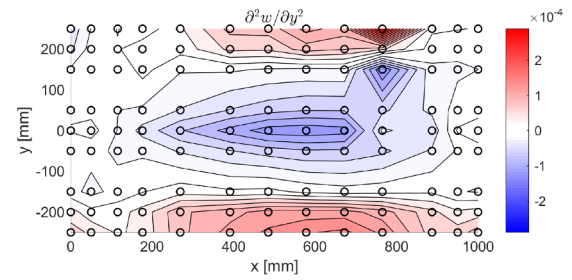
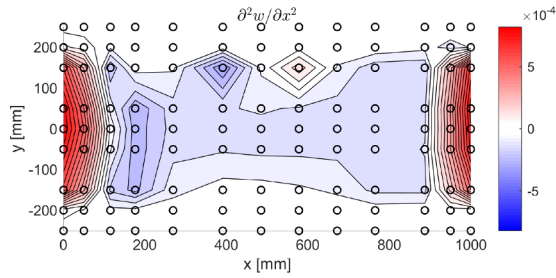
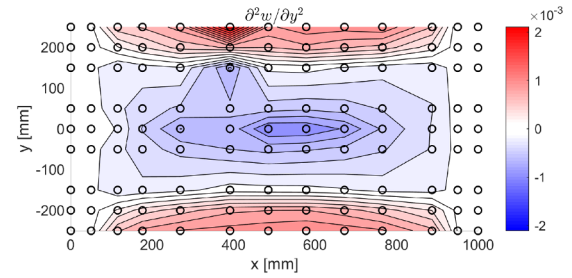
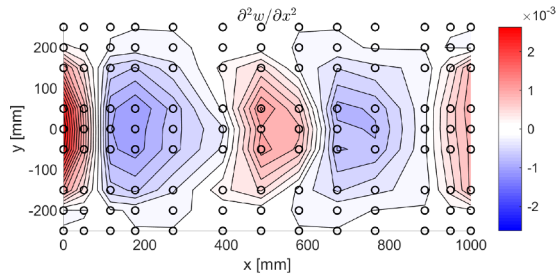
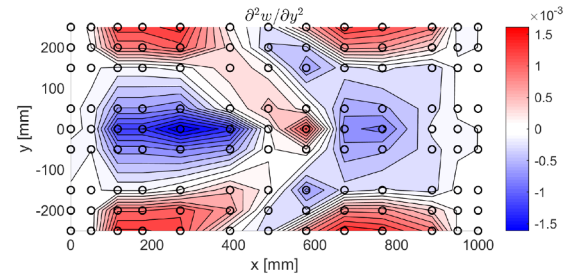
XX_10_30 condition, even though the general behaviour of the signals is about the same. On the contrary, for the 3 mm plate, the loss of correlation of the pressure signals is already evident at P_{12} , which means that the larger deformations caused by the increased loading have a strong effect on the pressure distribution. In the right column of Fig. 15 the time histories of pressure measured at the side are shown. No significant differences are observed for the 15 mm plate between the pressures measured in the middle and at the side. For the 3 mm plate the pressure at the side behaves similarly to that measured on the 15 mm plate, although for the 3 mm plate the pressure peaks are much



(a) 08_06_40.a



(b) 08_10_45.a

Fig. 12. Permanent bumps found on the 0.8 mm plates at the end of two tests. The trailing edge is on the left.(a) 3_10_30 $\partial^2 w / \partial x^2$ (b) 3_10_30 $\partial^2 w / \partial y^2$ (c) 08_04_45 $\partial^2 w / \partial x^2$ (d) 08_04_45 $\partial^2 w / \partial y^2$ (e) 08_10_45 $\partial^2 w / \partial x^2$ (f) 08_10_45 $\partial^2 w / \partial y^2$ **Fig. 13.** Second derivatives of the residual out-of-displacement fields in the x direction $\frac{\partial^2 w}{\partial x^2}$ (1/mm) on the left column and in the y direction $\frac{\partial^2 w}{\partial y^2}$ (1/mm) on the right column for the test conditions 3_10_30, 08_04_45 and 08_10_45. The trailing edge, corresponding to $x = 0$, is on the left. The black round circles indicate the positions of the measurement points..

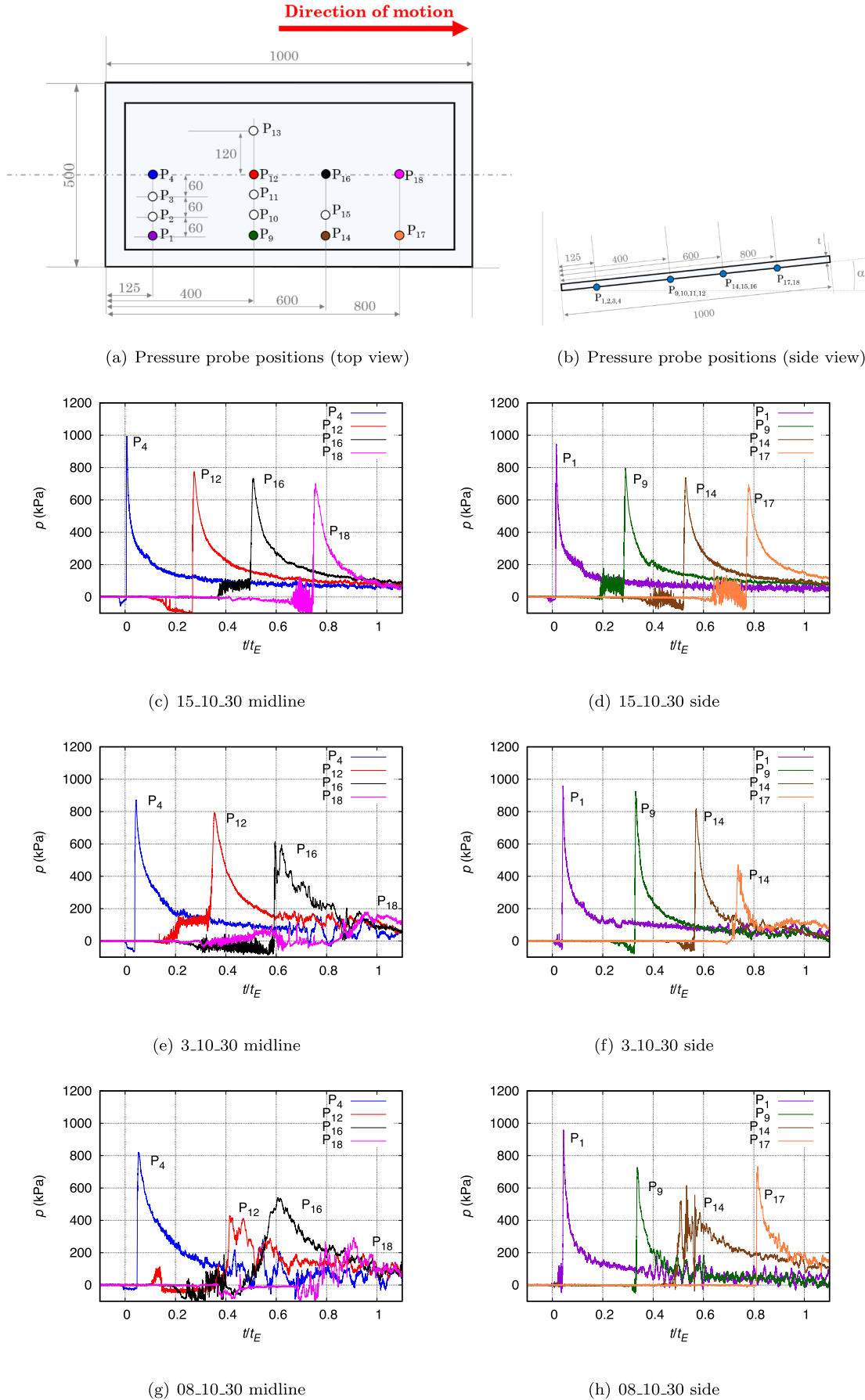


Fig. 14. Time histories recorded by the pressure probes located along the midline and at the side for the tests XX_10_30, i.e at $U = 30$ m/s, 10° pitch angle for the plate 15 mm thick (top) 3 mm thick (centre) and 0.8 mm thick (bottom).

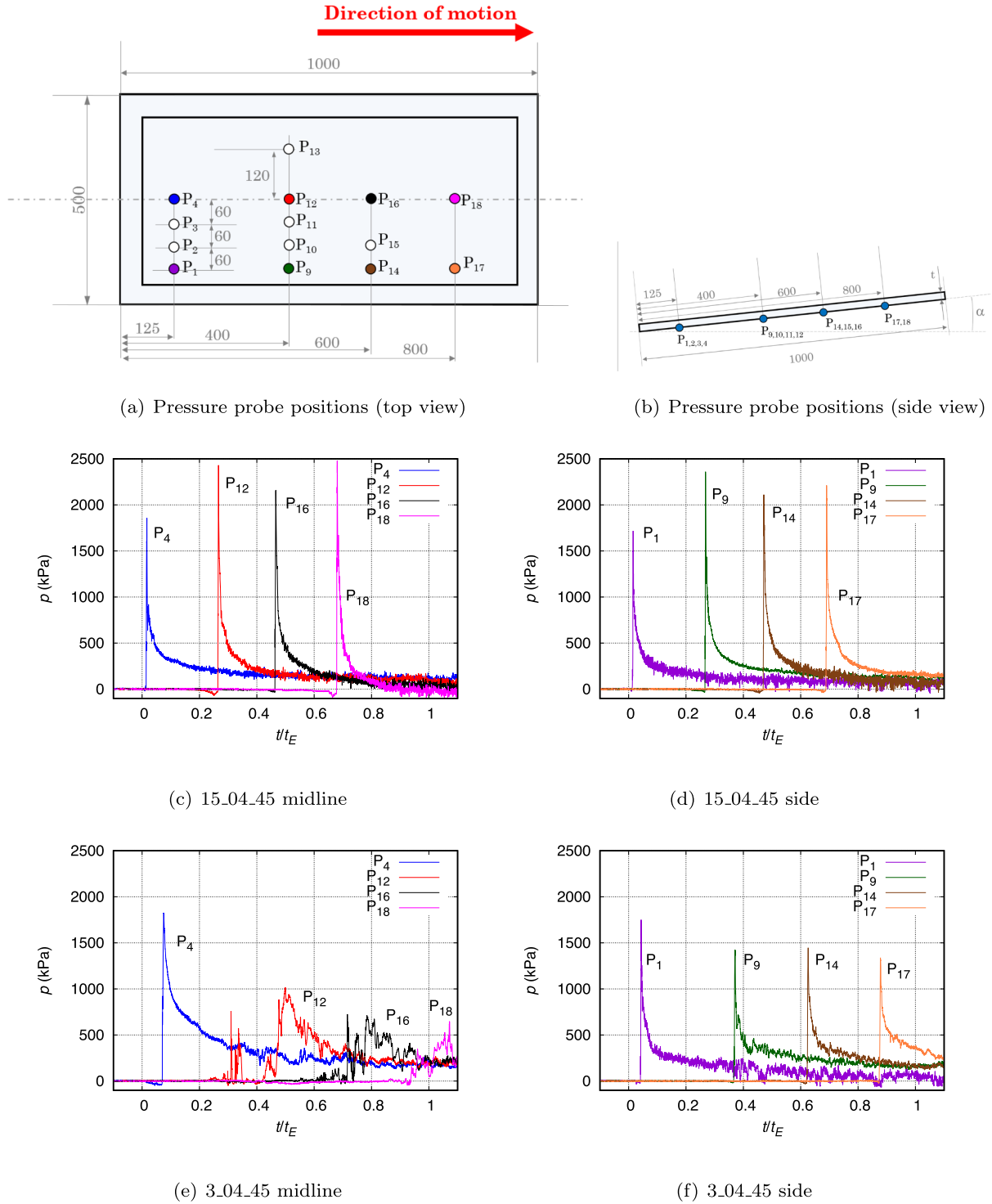


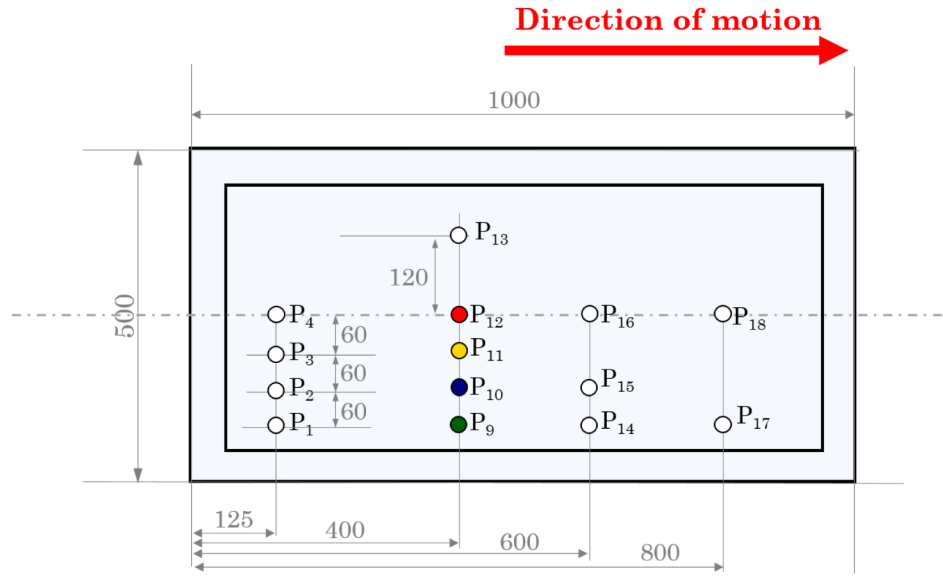
Fig. 15. Time histories recorded by the pressure probes located along the midline and at the side for the tests XX_04_45, i.e. at $U = 45$ m/s, 4° pitch angle for the plate 15 mm thick (top) and 3 mm thick (bottom).

lower. However, the pressure profiles in Fig. 15(f) are much more correlated than those in 15(e), thus indicating that the strongest fluid-structure interaction occurs in the middle of the plate.

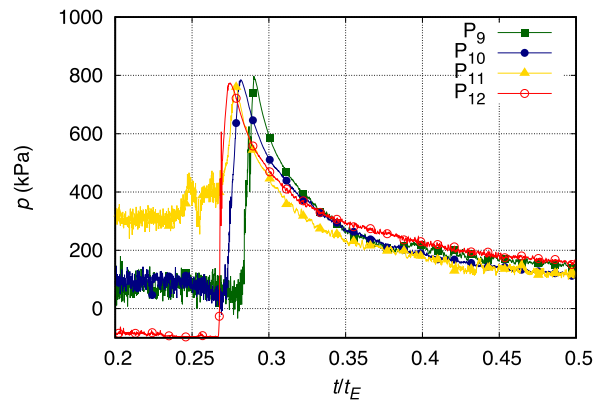
3.3.2. Pressure changes in the transverse direction

In Fig. 16 the time histories of the pressure measured in the tests XX_10_30, i.e. at 30 m/s, 10° pitch angle, by the probes located at 0.4 m from the trailing edge are shown. It is possible to spot a clear different trend between the thick and the thin plates. In fact, in the thick plate

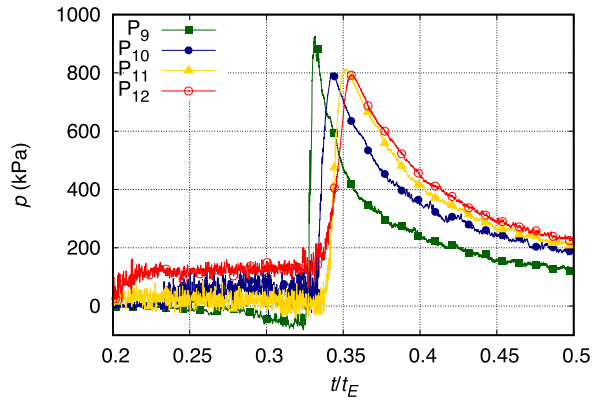
the pressure peak first reaches the probe in the middle, i.e. P_{12} , and next the other probes going from the midline to the side i.e. P_{11} , P_{10} and P_9 in succession. Therefore, it is inferred that the spray root has a backward curvature, as widely discussed in [33]. In the case of thin plates the pressure peak occurs at P_9 first and then, in succession, at the probes P_{10} , P_{11} and P_{12} , i.e. going from the side to the midline, thus denoting a forward curvature. Furthermore, it can be noticed that the occurrence of the first pressure peak in the thin plates is delayed compared to the thick plates.



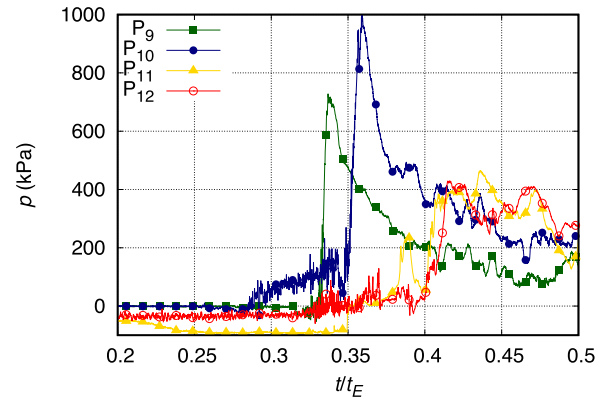
(a) Pressure probe positions



(b) 15_10_30 (thickness 15 mm)



(c) 3_10_30 (thickness 3 mm)



(d) 08_10_30 (thickness 0.8 mm)

Fig. 16. Time histories recorded by the pressure probes located at 0.4 m ahead of the trailing edge for the tests XX_10_30, i.e. at $U = 30$ m/s, 10° pitch angle for the 15 mm plates, the 3 mm plate and the 0.8 mm plate.

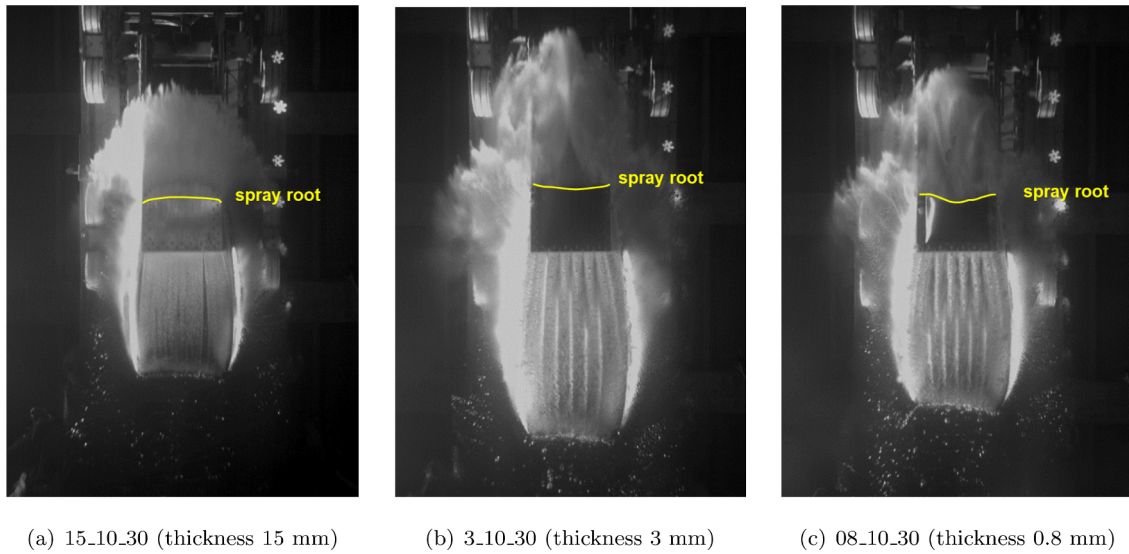


Fig. 17. Underwater images at the instant in which the spray root is at the location of the probe P_{12} : test conditions XX_10_30, i.e. $U = 30$ m/s, $\alpha = 10^\circ$ and thickness 15 mm (a), 3 mm (b) and 0.8 mm (c).

The changes in the curvature of the spray root and the air entrapment in the central part of the plate during the impact phase can also be noticed in the underwater images, as displayed in Fig. 17, where the spray root line is highlighted by the yellow line. In that figure, in the case of the thick plate, the spray root line exhibits a backward curvature. In the case of thin plates, a forward curvature of the spray root is observed, which is more pronounced for the 0.8 mm plate. In the latter case, the shape of the spray root is strongly affected by the structural deformation of the plate, which is the reason of the sharp curvature in the profile near the inner frame. The light reflections from the sides, particularly evident in Fig. 17(c), clearly highlight the large out-of-plate deformation caused by the impact.

The changes in the spray root curvature can also be noticed in the tests at 45 m/s, pitch angle 4° , as shown in Fig. 18. Also in this case, the sequence of the pressure peaks for the 3 mm plate is reversed with respect to that observed for the 15 mm plate. Moreover, for the 3 mm plate the pressure peaks exhibit an increased spreading in time and their values are lower.

Again, looking at the underwater images in Fig. 19, it is observed that the spray root curvature in the 3 mm case is the opposite of the one observed for the thick plate, and it is much higher than that observed for the cases XX_10_30, due to the higher loading and to the more significant fluid-structure interaction. The effect is even more evident in the 0.8 mm plate. The large out-of-plane deformation is responsible for the spreading of the pressure peaks in both space and time. The underwater images also highlight the formation of two jets at the sides converging towards the midline.

3.4. Combined analysis of pressure and strain measurements

It is worth analysing simultaneously pressures and strains in order to better understand the relationship between the structural deformation and the hydrodynamic loading, as well as their mutual interaction.

To this purpose, the pressures and the x-strains recorded along the midline during the impact phase for the tests at 30 m/s, 10° pitch angle and different thickness are shown in Fig. 20. In the case of the thick plate, there is a quite good correlation between the pressures and the x-strains, with the strains attaining a maximum concurrently with the passage of the pressure peak beneath. After the passage of the peak, the strains gradually diminish and approach zero, as time elapses. In the 3 mm plate, the x-strain still exhibits a maximum concurrent with the passage of the pressure peak, but in this case the pulse width is

narrower and less pronounced. Furthermore, due to the reduced thickness of the plate, the structural deformation propagates much further ahead of the spray root and all the strains measured by the gauges, except for S_{1x} , attain about the same strain level when reached by the pressure peak and remain quite flat afterwards, until the end of the impact phase. Owing to the disruption of the spray root and to the spreading of the pressure peak, there is no evident maximum for S_{5x} , which also approaches the strain values reached by the other gauges. Of course the strain levels are much higher than in the 15 mm case. In the 0.8 mm plate, a clear peak is seen for S_{1x} and, to a lesser extent, for S_{7x} . This is a consequence of the peaked pressure distribution observed in the early stage of the impact, e.g. P_4 . Moving further, the pressure is spread over a wider area, which is the reason why, rather than a clear peak, the strains exhibit an almost flat region at high values and diminish afterwards. This is the case of S_{2x} for $t/t_E \in (0.3, 0.7)$, S_{8x} for $t/t_E \in (0.6, 0.8)$ and S_{5x} for $t/t_E \in (0.7, 0.9)$. Differently from the 3 mm case, the strain levels vary substantially from gauge to gauge. Moreover, the reduced thickness and the complicated flow developing beneath the plate induce visible oscillations in the strains.

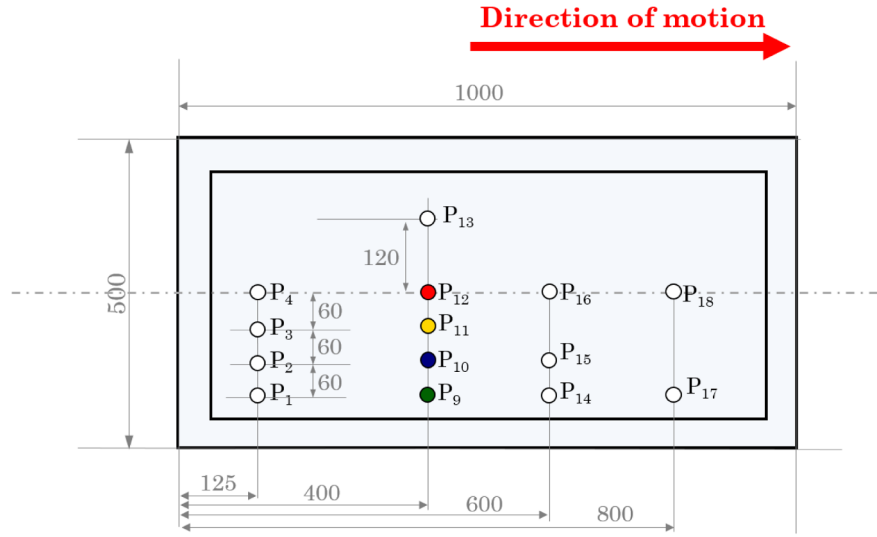
Similar conclusions can be drawn by looking at the data of the tests performed at 45 m/s, 4° pitch angle for the 15 mm and 3 mm plates, shown in Fig. 21. At least for the 15 mm plate, owing to the smaller pitch angle, the pressure peak is much sharper than that observed in test condition 15_10_30, see Fig. 20(b), and there is a quite high correlation between the pressure and the strain profiles. Similarly to what found in the test condition 3_10_30, for the 3 mm plates the strains grow gradually and the values of all gauges approach about the same level, even though this occurs at $t/t_E \approx 0.8$ in the case 3_04_45 and at $t/t_E \approx 0.6$ in the case 3_10_30, see Fig. 20(b).

3.5. Hydrodynamics loads

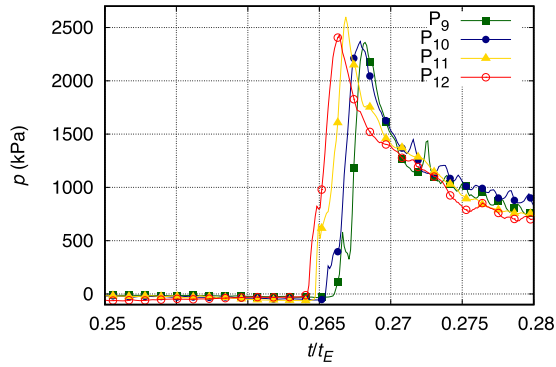
As already noticed in [32], the tangential load F_x is not so significant during the impact phase. F_x undergoes a sharp rise as soon as the spray root reaches the leading edge of the plate, and the front face of the acquisition box drops down to the still water level.

Much more relevant is the normal force F_z ². In Fig. 22 comparisons between the time histories of the z-force measured in the tests XX_10_30 (i.e. $U = 30$ m/s and $\alpha = 10^\circ$), XX_06_40 ($U = 40$ m/s and $\alpha = 6^\circ$) and

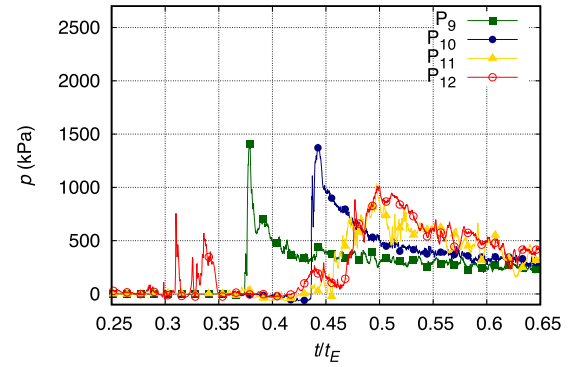
² The sign convention for F_z and F_x is consistent with the coordinate system shown in Fig. 1.



(a) Pressure probe positions

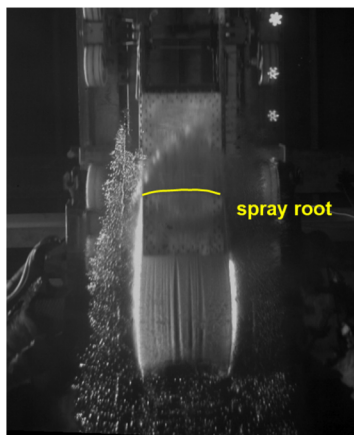


(b) 15.04.45 (thickness 15 mm)

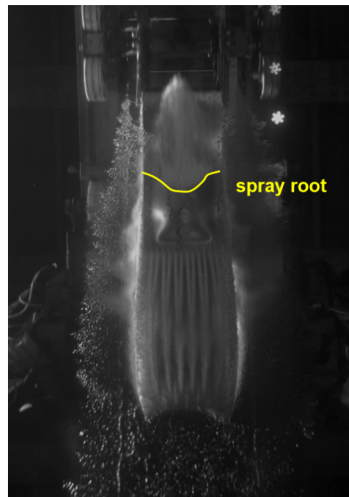


(c) 3.04.45 (thickness 3 mm)

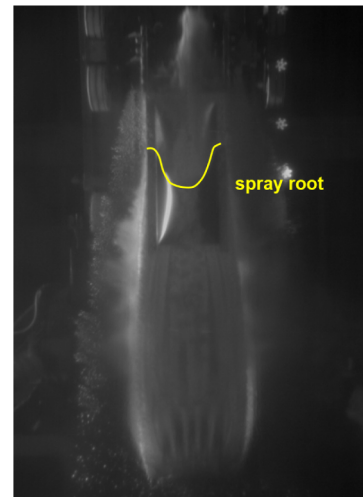
Fig. 18. Time histories recorded by the pressure probes located at 0.4 m ahead of the trailing edge of the plate for the test conditions XX_04_45, i.e. at $U = 45$ m/s, 4° pitch angle for the 15 mm plate and the 3 mm plate.



(a) 15.04.45 (thickness 15 mm)



(b) 3.04.45 (thickness 3 mm)



(c) 08.04.45 (thickness 0.8 mm)

Fig. 19. Underwater images at the instant in which the spray root is at the location of the probe P_{12} : test conditions XX_04_45, i.e. $U = 45$ m/s, $\alpha = 4^\circ$ and thickness 15 mm (a), 3 mm (b) and 0.8 mm (c).

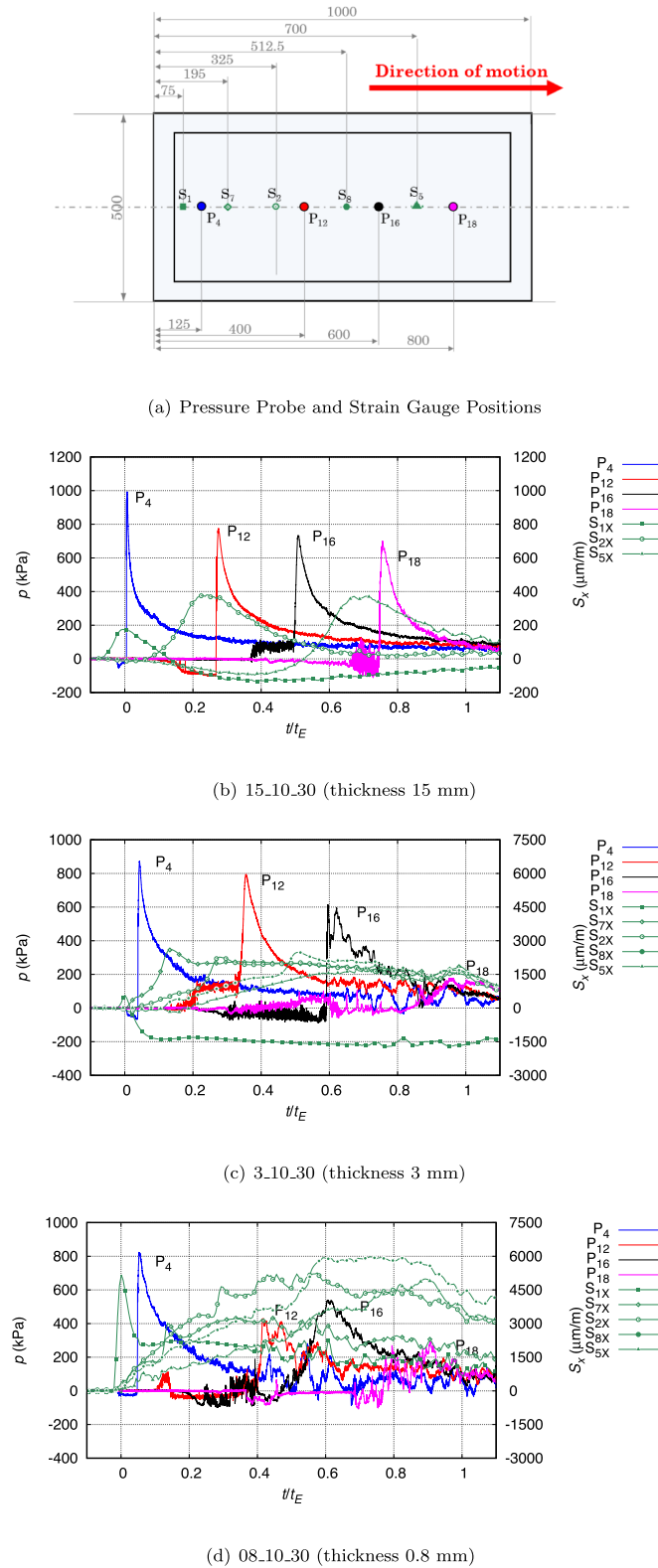


Fig. 20. Time histories of the pressures and longitudinal strains measured along the midline for the tests XX_10_30, i.e $U = 30$ m/s, 10° pitch angle for 15 mm plate, the 3 mm plate and the 0.8 mm plate.

XX_04_45 ($U = 45$ m/s and $\alpha = 4^\circ$) are shown. It is worth reminding that the rise of the z -force occurs slightly earlier than at $t = 0$, because the origin of time is chosen to be at the peak of S_{1x} , which is located slightly ahead of the trailing edge, as explained in Section 2. In all cases there is

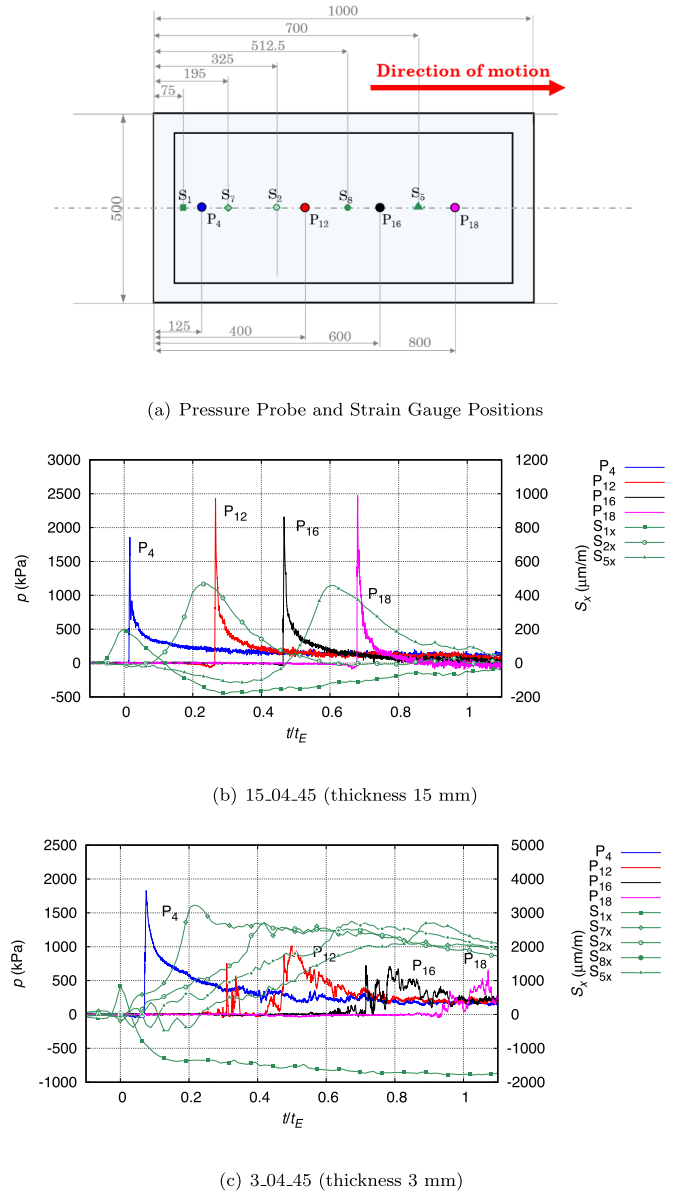


Fig. 21. Time histories of the pressures and longitudinal strains measured along the midline for the tests XX_04_45, i.e $U = 45$ m/s, 4° pitch angle for 15 mm plate and the 3 mm plate.

an initial stage, lasting up to $t/t_E = 0.2$, during which the normal load is almost independent of the plate thickness. This indicates that, even for the thinnest plates, the out-of-plane deformation in the early stage is not large enough to affect the loading. However, in the following stage the curves start to diverge, with the normal load increasing when reducing the plate thickness, further confirming that the deformation of the plate affects the hydrodynamics beneath the plate. Based on the above discussion on the behaviour of the pressure, even though the intensity of the pressure peak drops substantially as the structural deformation increases, presumably the integrated pressure along the plate is higher. This result is consistent with what found numerically in [11], where it is shown that the loading increases with the structural deformation. It is also worth remarking that the numerical results in [11], obtained with two different numerical approaches, show a very good agreement with the present experimental results for the 15 mm plate, the 3 mm plate and the 0.8 mm plate at 6° pitch angle and at a horizontal speed of 40 m/s. From a quantitative viewpoint when reducing the plate stiffness, the normal load increases, and the maximum values

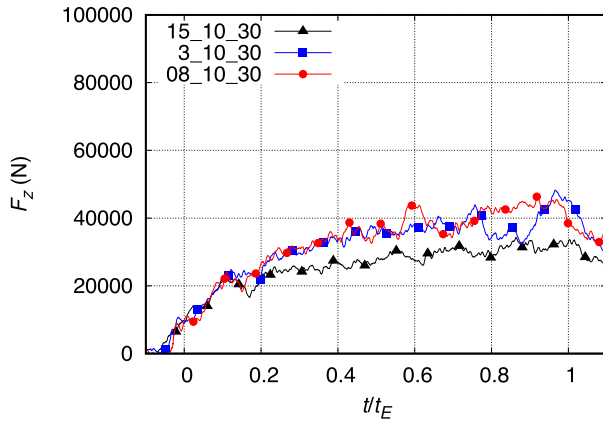
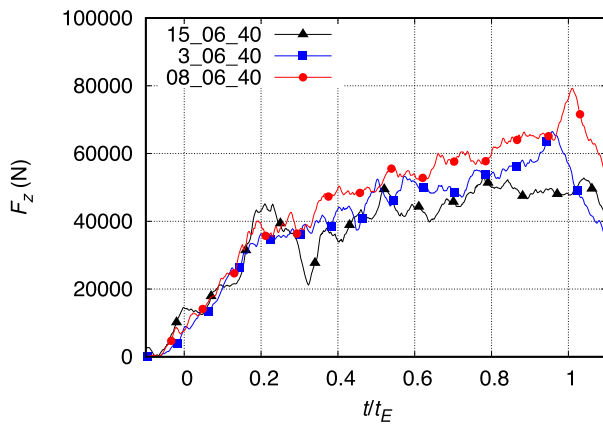
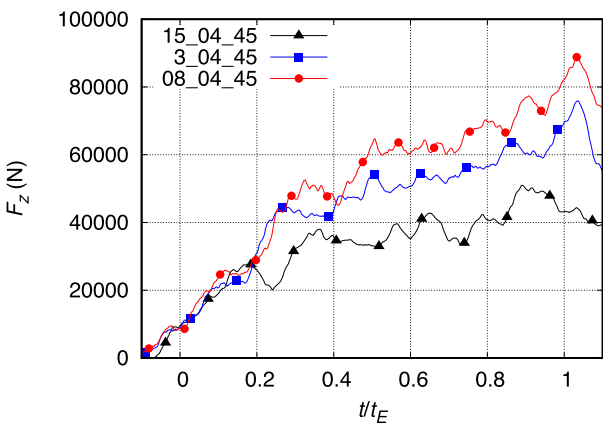
(a) XX_10_30 - $U = 30$ m/s and $\alpha = 10^\circ$ (b) XX_06_40 - $U = 40$ m/s and $\alpha = 6^\circ$ (c) XX_04_45 - $U = 45$ m/s and $\alpha = 4^\circ$

Fig. 22. Time histories of the normal force at different thickness for the test conditions XX_10_30, XX_06_40 and XX_04_45.

are from 30 to 50% higher than those measured on the thick plate.

It is worth observing that the difference between the normal loading grows when increasing the horizontal speed and reducing the pitch angle. This result is valid for the thick plates, see [32,33], and it is also observed in the thin plates, as shown for example in Fig. 23, where the time histories of the normal load acting on the 0.8 mm plate at different

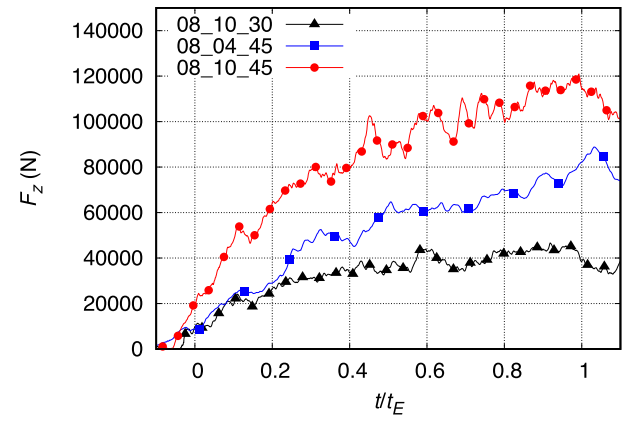


Fig. 23. Time histories of the normal force for the 0.8 mm plate at different horizontal speeds and pitch angles.

pitch angles and horizontal speed are shown. In [33] the normal loading measured for the thick plate were found to scale with the projection of the force exerted by the stagnation pressure $\rho(U^2 + V^2)/2$, assumed to be uniformly distributed all over the plate surface, onto the direction normal to the plate.³ The same scaling is adopted here, by drawing in Fig. 24 the quantity $F_s = F_z/[\rho(U^2 + V^2)\sin(\alpha + \beta)]$. Each plot shows F_s at a different plate thickness, but at a different horizontal velocity and pitch angle. It can be observed that for the thick plates (15 mm) the time history of F_s are fairly overlapped, with a limited scatter, thus confirming the validity of the scaling, which however no longer holds for the thin plates. In fact, for the thin plates, there is an initial phase, up to $t/t_E = 0.15 - 0.2$ in which the time histories of the scaled loads collapse. Then, as the plate penetrates water, the time histories start to diverge significantly, as a consequence of the plate deformation and of the fluid-structure interaction. The scaled values grow when increasing the test speed and the pitch angle. Since the increase in both parameters leads to an increase in the plate deformation, the results again indicate that the deformation amplifies the loading and, furthermore, that such amplification does not follow the scaling. Clearly these effects are more significant for the thinnest plates (0.8 mm), where the deformations are larger.

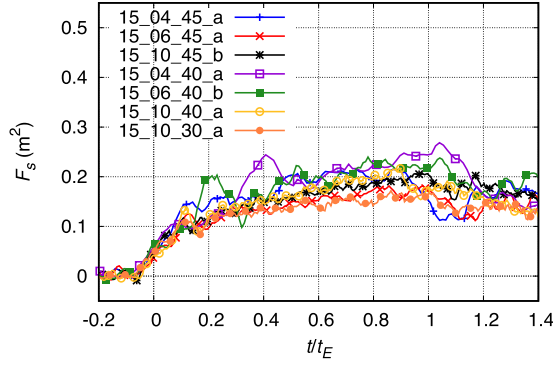
The significant effect of the structural deformation on the loading and the difficulties in the scaling have important consequences. First of all, it is not always true that tests on rigid specimens provide conservative estimates of the loading. Secondly, owing to the difficulties in reproducing the correct scaling of the structure and the changes in the loads induced by the structural deformation, scaled model tests of deformable structures cannot be considered as completely reliable, and a verification at full-scale conditions, i.e. at the ditching conditions and on a structure representative of a real aircraft (especially with the same the panel thickness), is advisable, whenever feasible.

By using the data of the forces provided by the rear and forward load cells, it is possible to retrieve an estimate of the position of the centre of loads during the impact phase. This is shown in Fig. 25. The centre of loads is provided as its distance from the trailing edge x_{CL} measured along the plate, as indicated in Fig. 1, and it is given by:

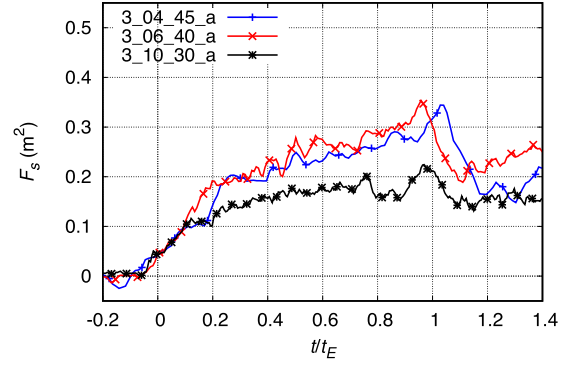
$$x_{CL} = x_0 + \frac{L F_{zF}}{F_{zR} + F_{zF}}$$

where L is the distance between the two load cells (1.385 m), x_0 is the distance between the trailing edge and the rear load cell (0.02 m) and F_{zF} and F_{zR} are the forces measured by the forward and rear load cell respectively. The position of the centre of loads at a certain time instant is of course related to the pressure distribution over the plate at that

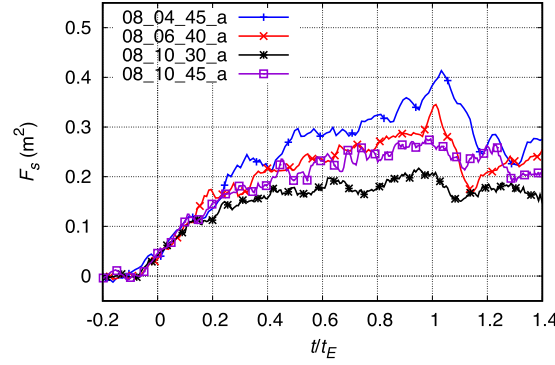
³ In [33] there was a mistake in the text and the data were scaled by $\rho(U^2 + V^2)\sin(\alpha + \beta)$ and not by ρU_n^2 as indicated in the captions.



(a) 15_XX_YY - 15 mm



(b) 3_XX_YY - 3 mm



(c) 08_XX_YY - 0.8 mm

Fig. 24. Time histories of the scaled normal force for different test conditions.

instant. As the plate penetrates water, the centre of load moves at a certain speed towards the leading edge, as a consequence of the propagation of the spray root and of the corresponding pressure peak. At the end of the impact phase the pressure peak reaches the leading edge and leaves the plate surface, hence the centre of loads moves backwards. For the cases XX_10_30 i.e. at $U = 30$ m/s and pitch angle 10° , the structural deformations are not so relevant, thus there are no substantial changes in the position of the centre of loads. This means that, even if the total loads tend to slightly increase as the thickness decreases, the pressure distribution at each time instant is not particularly affected by the fluid-structure interaction. For the test conditions XX_06_40 and XX_04_45, instead, the centre of loads for the thin plate propagates more slowly towards the leading edge compared to the thick plates. This is because, due to the deformation of the plate, the spray root propagation velocity is reduced, particularly in the middle of the plate, as shown in Figs. 15 and 19, and, correspondingly the associated pressure distribution is more spread compared to the case of the thick plates. Owing to the increased loading, the reduction in the propagation velocity of the centre of loads is even more evident for the test condition XX_04_45 i.e. at 45 m/s and at 4° pitch angle, as shown in Fig. 25(c).

4. Conclusions

In this paper an analysis of the fluid-structure interaction phenomena occurring during the water impact of aluminium plates of different thickness at high horizontal velocity has been carried out based on experimental measurements. Test conditions are, as much as possible, close to those of real aircraft ditching scenarios.

It has been found that the reduction of the thickness (and so of the stiffness) of the plate induces a higher level of strains, which overcome the yielding limit in some cases. The structural behaviour changes substantially due to both the different stiffness and to the fluid-structure interaction, which leads to a quite different hydrodynamic loading.

A careful analysis of the pressure and strains has revealed that the correlation between them is progressively reduced as the structural deformation increases with the reduction of the stiffness as well as with the penetration of the plate.

By looking at the time histories of pressure, a general reduction of the pressure peaks is observed, which is accompanied by a spreading of the pressure distribution in both space and time. Particularly evident is the effect of the structural deformation on the propagation of the spray

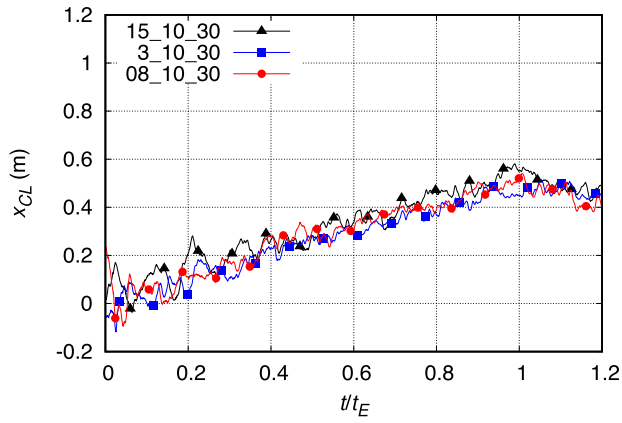
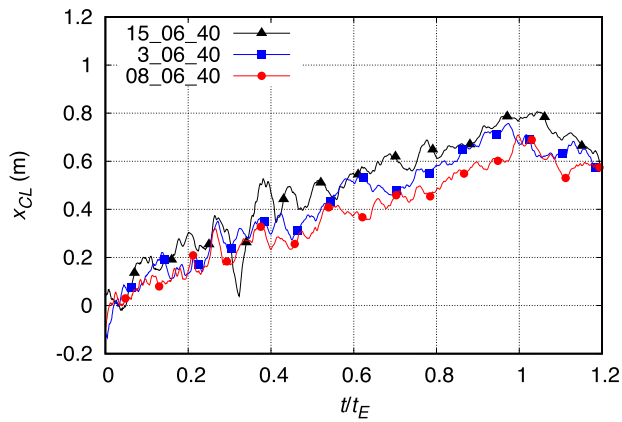
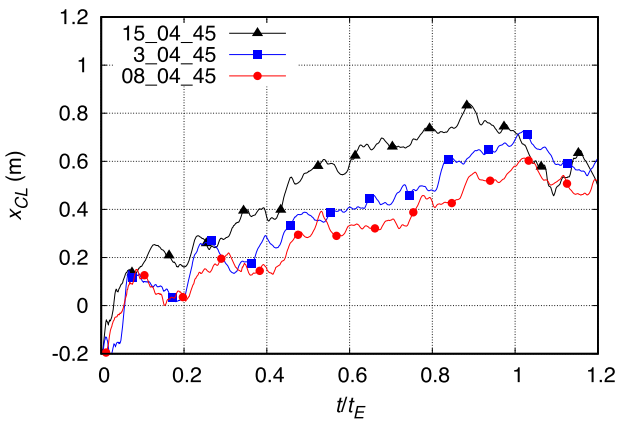
(a) XX_10_30 - $U = 30$ m/s and $\alpha = 10^\circ$ (b) XX_06_40 - $U = 40$ m/s and $\alpha = 6^\circ$ (c) XX_04_45 - $U = 45$ m/s and $\alpha = 4^\circ$

Fig. 25. Position of the centre of loads during the impact phase at different thickness for the conditions XX_10_30, XX_06_40 and XX_04_45.

root. The pressure peak/spray root propagation speed is observed to be reduced for the thin plates compared to thick plates. It is also found that

Appendix A. Ratio of the wetting time to the first natural period of the plate

As noticed in Faltinsen [20,37], the relevance of the fluid-structure interaction can be indicated by the ratio of the wetting time to the first wetted natural period T_{11} (related to the mode 1-1) of the flat plate.

the out-of-plane deformation leads to an inversion of the curvature of the spray root line compared to the case of the thick plates, as also attested by the underwater images.

An increase in the total loading with the increase in the structural deformation is observed. Such increase is noticed starting from about $t/t_E \approx 0.2$ (i.e. from 1/5th of the impact phase) and progressively grows up to the end of the impact phase, when it can be 30% to 50% higher than the corresponding loading in the thick plates. Differently from the thick plates, the scaling with the normal projection of the force exerted by the stagnation pressure is no longer valid in presence of large structural deformation, i.e. for $t/t_E > 0.2$.

Two practical conclusions can be drawn based on the above considerations. First of all, differently from what it is usually assumed and even without a strong hydroelastic coupling, tests on rigid structures do not always provide conservative estimates of the loads. Such a circumstance depends on the details of the structure and on the hydrodynamic loading, and it is difficult to judge in advance. Secondly, in presence of a strong fluid-structure interaction, such as that observed in these tests, the hydrodynamic loads are strongly dependent on the structural behaviour, and it is impossible to determine a proper scaling that works for geometry, velocity, loads and structure at the same time.

The above considerations pave the way for future and more focused investigations on the subject. First of all, a deeper comprehension of the strong fluid-structure interaction phenomena and, specifically, of the way in which the loading changes as a consequence of the structural deformation is deemed essential. Furthermore, such studies could also help to understand whether there is a way to scale the structure that can also account for the variation of the loads associated with the plate deformation. Finally, it would be interesting to analyse what happens in terms of the energy balance and the energy exchange between the fluid and the structure. Owing to the complexity of the problem and to the need of achieving a detailed characterization of the flow field, of the pressure distribution and of the resulting structural loading and deformation, such studies can only be addressed by high-fidelity, fully coupled, fluid-structural numerical models. To this purpose, the unique data coming from this rather extensive experimental campaign surely constitute a solid database for the verification and validation of such numerical tools.

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CRediT authorship contribution statement

Emanuele Spinosa: Formal analysis, Methodology, Conceptualization, Visualization, Writing - original draft, Writing - review & editing. **Alessandro Iafrati:** Conceptualization, Investigation, Data curation, Methodology, Resources, Supervision, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The wetting time can be estimated on first approximation, as the duration of the impact phase t_E . The first wetted natural period, instead, can be derived using a simplified approach, which is described hereafter.

In the following a , b and h are the length, the width and the thickness of the plate respectively. Since the outer part of the plate is reinforced by the frame, it is not allowed to deform, therefore it is assumed that the values of a and b are slightly reduced, being 0.9 m by 0.4 m respectively, instead of 1 m by 0.5 m

As anticipated in Section 2, the boundary condition on the edges of the plate can be considered a quasi-clamped boundary condition, in between a simply-supported condition at all sides (SS) and a clamped condition at all sides (C). This has been verified a posteriori by comparing the results of numerical modal FEM simulations and experimental modal tests, which were performed on the 3 mm plate. Given that, both these conditions are examined in this procedure.

In the case of simply supported plate at all sides (SS) the first natural (dry) frequency can be estimated using the relationship (4.20) in Leissa [38]:

$$f_{11}^{SS} = \frac{1}{2\pi} \sqrt{\frac{D}{\rho'_m} \left[\left(\frac{\pi}{a} \right)^2 + \left(\frac{\pi}{b} \right)^2 \right]} \quad (\text{A.1})$$

In the previous relationships $D = \frac{E h^3}{12(1-\nu^2)}$, E , ν and ρ'_m are the Young modulus, the Poisson's ratio and the mass of the material per unit area respectively. For the plate material, the aluminium alloy AL2024-T3, $E = 73.1$ GPa, $\nu = 0.33$ and $\rho'_m = \rho_m h$, where ρ_m is the density of the material i.e. 2780 kg/m³.

In the case of clamped edges at all sides (C) the first natural frequency can be determined using the relationship

$$f_{11}^C = \frac{1}{2\pi} \sqrt{\frac{D}{\rho'_m} \frac{K_C}{b^2}}, \quad (\text{A.2})$$

where K_C is the value displayed in Table 4.28 (first row) in [38]. K_C depends on the ratio a/b , which is 2.25 in the present case. In [38] it is shown that K_C is 24.56 for $a/b = 2$ and 23.76 for $a/b = 2.5$. It is reasonable to assume that, for $a/b = 2.25$, K_C is the average of the two values reported above, hence $K_C = 24.16$.

To estimate the wetted natural frequency/period, it is assumed that the mass of the material per unit area is increased by an amount ρ'_a , which is the added mass per unit area of a flat plate initially floating on the free surface and moving suddenly downwards (heave). On first approximation, the total added mass of the plate can be estimated as outlined in Newman [39]. In particular, in Table 4.3 the value of the added mass of a flat plate of unit length and half-width b that moves in a vertical direction in an unbounded domain is provided. In the case of plate impact, the added mass at heave, which we denote as m_a , is halved (see Newman [39], Section 6.17), hence we obtain:

$$m_a = \frac{1}{8} \pi \rho a b^2 \quad (\text{A.3})$$

where ρ is the water density. Assuming that the added mass is distributed uniformly over the plate surface, the value of ρ'_a , the added mass per unit surface, is given by:

$$\rho'_a = \frac{m_a}{a b} = \frac{\pi}{8} \rho b \quad (\text{A.4})$$

The first natural wetted frequency can then be estimated using the relationships (A.1) and (A.2), but replacing ρ'_m with $\rho'_w \doteq \rho'_m + \rho'_a$:

$$f_{11}^{SS} = \frac{1}{2\pi} \sqrt{\frac{D}{\rho'_w} \left[\left(\frac{\pi}{a} \right)^2 + \left(\frac{\pi}{b} \right)^2 \right]} \quad f_{11}^C = \frac{1}{2\pi} \sqrt{\frac{D}{\rho'_w} \frac{24.16}{b^2}}. \quad (\text{A.5})$$

For the quasi-clamped boundary condition, it can be assumed that the first natural wetted frequency is the average of these two values, i.e.

$$f_{11}^{wet} = 1/2 (f_{11}^{SS} + f_{11}^C) \quad (\text{A.6})$$

The ratio t_E/T_{11} is shown in Table A.2 for a combination of three plate thickness values, namely 15 mm, 3 mm and 0.8 mm, and three pitch angles, namely $\alpha = 4^\circ$, $\alpha = 6^\circ$ and $\alpha = 10^\circ$. It is worth observing that the values of t_E in this case are not determined as described in Section 2, but on first approximation as $t_E = L \sin \alpha / V$, which corresponds to the assumption that the flat plate does not deform significantly during the water entry.

Faltinsen [20] reports that for a ratio t_E/T_{11} higher than 1 a quasi-static approach can be applied to the transient problem by assuming the structure as rigid, whereas for $t_E/T_{11} \leq 0.25$ a coupling between the hydrodynamic and the elastic problem is expected to occur. Based on the data shown in Table A.2, a significant fluid-structure interaction is expected for the 0.8 mm plate at all pitch angles, but it could also be relevant for the 3 mm plate, especially at $\alpha = 4^\circ$ and $\alpha = 6^\circ$.

Table A.2

Ratio of the wetting time to the first wetted natural period of the flat plates for the test conditions examined.

t_E/T_{11}	$\alpha = 4^\circ$	$\alpha = 6^\circ$	$\alpha = 10^\circ$
$t = 15$ mm	8.97	13.4	22.3
$t = 3$ mm	0.88	1.31	2.18
$t = 0.8$ mm	0.12	0.18	0.31

Appendix B. Test repeatability

In order to assess the repeatability of the test conditions, the time histories of the strains S_1 , S_2 , S_3 and S_5 in the longitudinal (S_x) and transverse (S_y) directions for the tests performed on the 0.8 mm plate tested at 40 m/s and 6° are shown in Fig. B.26. In the figures, the time t_E of each test is indicated with a vertical line of the same colour as that of the corresponding line plot.

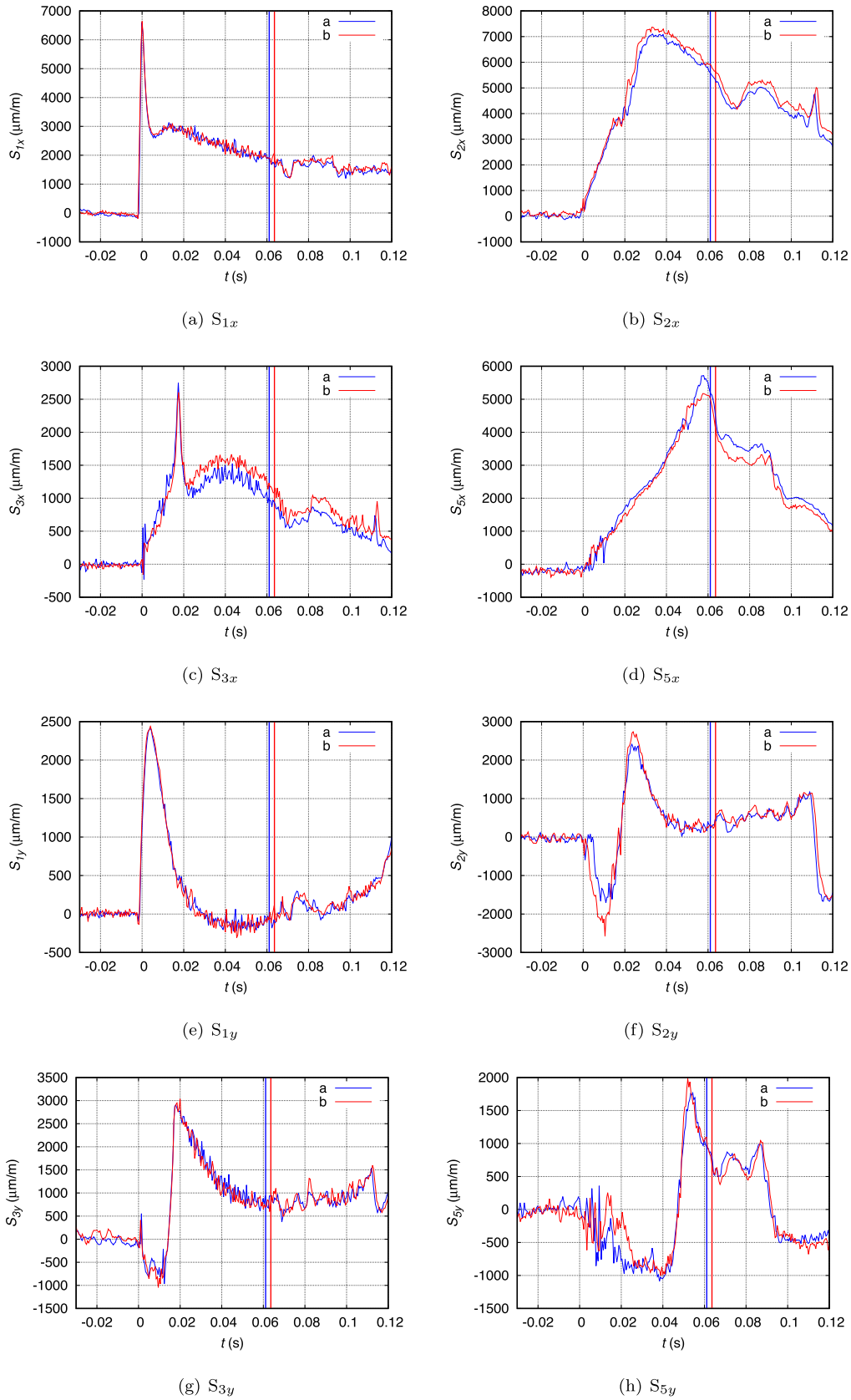


Fig. B.26. Comparison of the longitudinal and transverse strains recorded by the strain gauges S_1 , S_2 , S_3 and S_5 for the 0.8 mm plate at $U = 40$ m/s, pitch angle 6° for the two repeats.

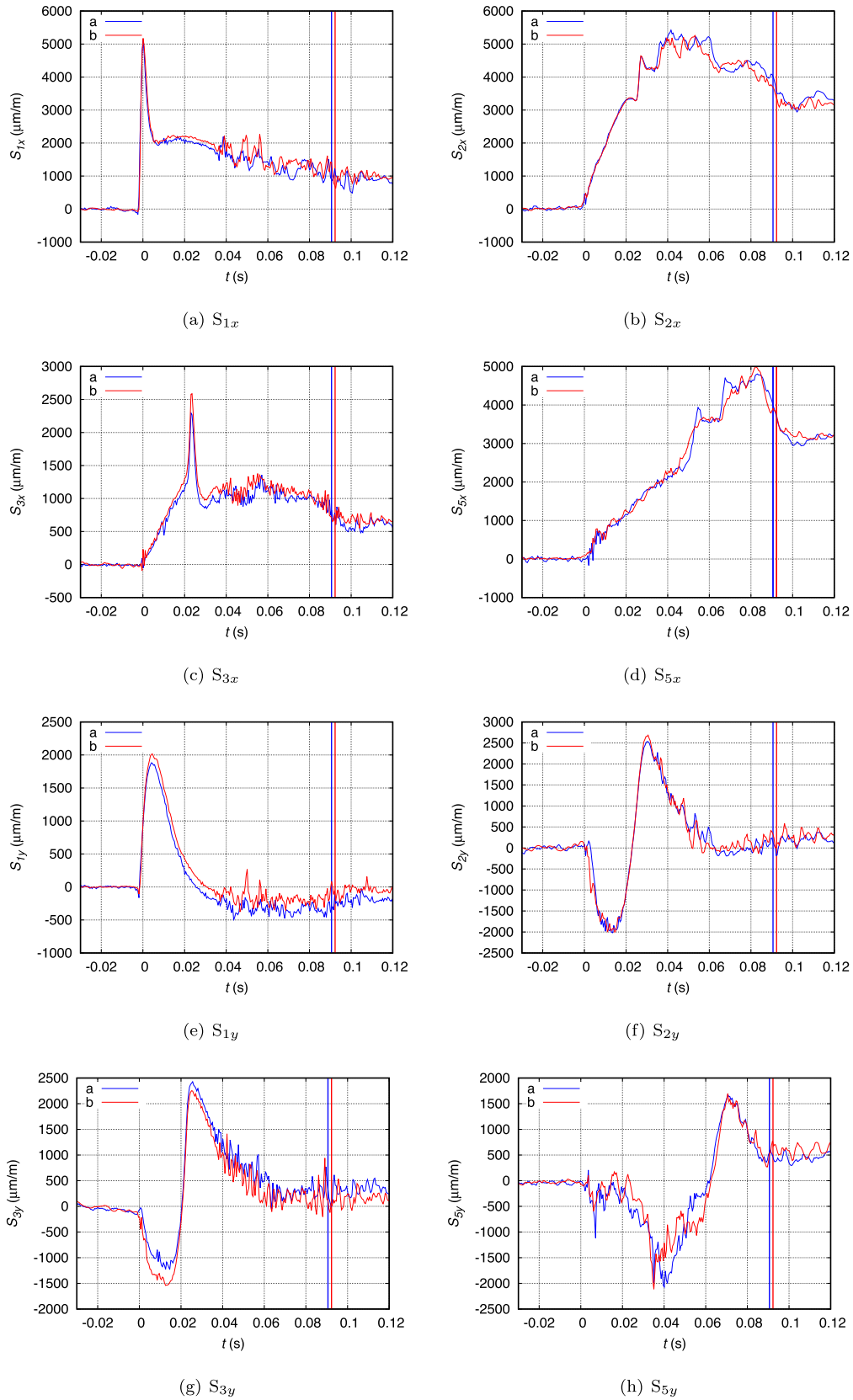


Fig. B.27. Comparison of the longitudinal and transverse strains recorded by the strain gauges S_1 , S_2 , S_3 and S_5 for the 0.8 mm plate at $U = 30$ m/s, pitch angle 10° for the two repeats.

A good repeatability is observed for the strain gauges located along the midline, i.e. S_1 , S_2 and S_5 , as well as for the strain gauge located at the side, i.e. S_3 , for which a higher sensitivity to the boundary conditions is expected.

A similar comparison is established in Fig. B.27 for the test at $U = 30$ m/s, thickness 0.8 mm and 10° pitch angle. In this case, pressure probes were

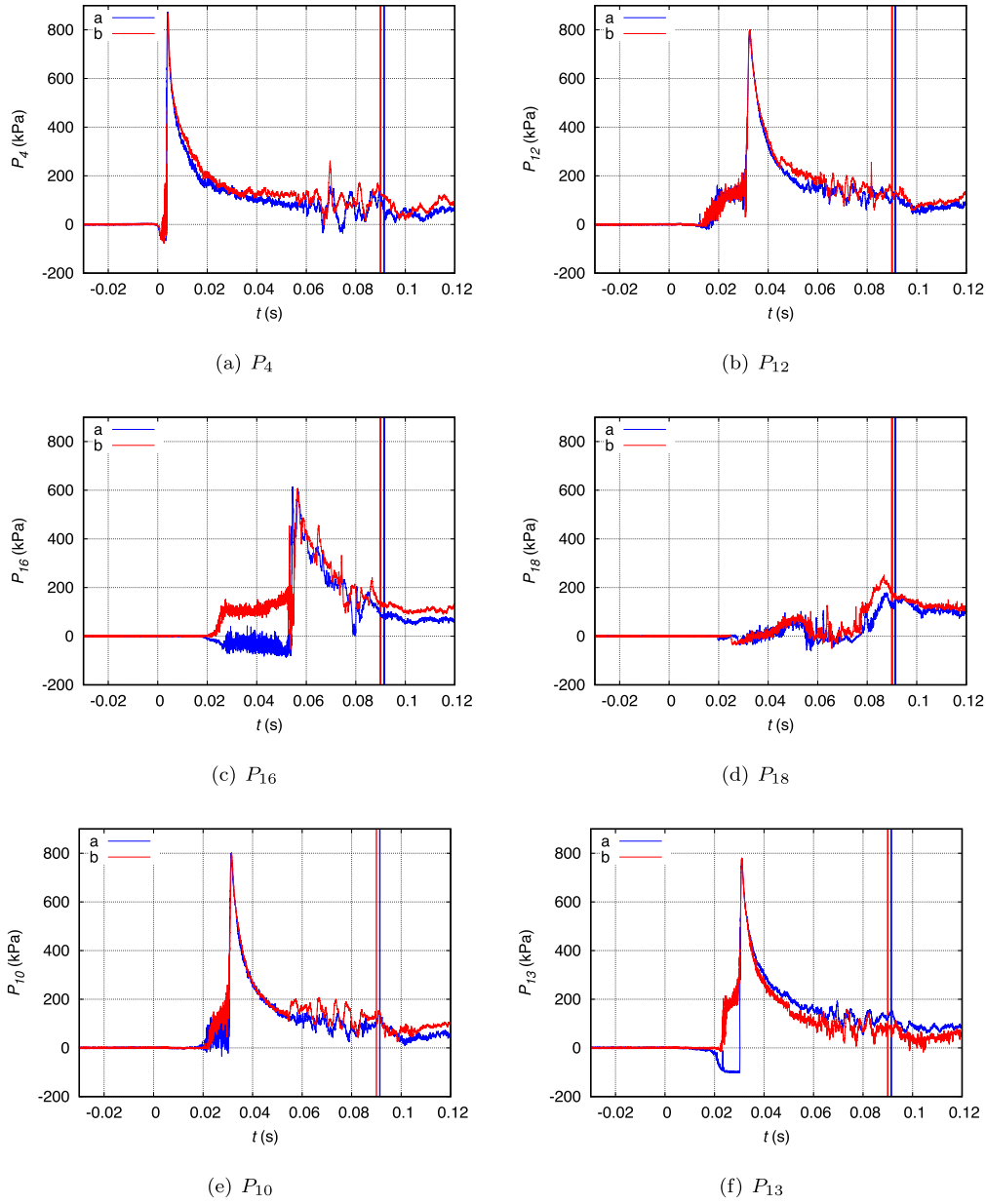


Fig. B.28. Comparison of pressure time histories for the test condition 3_10_30, i.e. at $U = 30$ m/s, thickness 3 mm and pitch angle 10° for the two repeats.

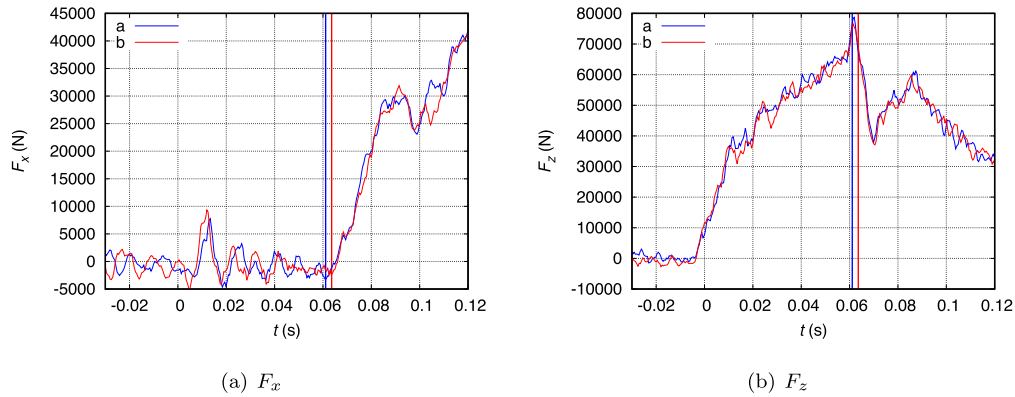


Fig. B.29. Comparison of the total forces for the test condition 08_06_40, i.e. at $U = 40$ m/s, thickness 0.8 mm and pitch angle 6° for the two repeats.

installed on one plate only. The good agreement of the data enables to say that there are no significant changes in terms of the structural behaviour induced neither by the presence of the holes drilled to install the pressure probes, nor by the pressure probes themselves.

As for the pressure measurements, the repeatability is assessed for the tests performed at 30 m/s, 10° pitch angle, for the 3 mm plate (Fig. B.28).

Comparisons are shown for the probes located along the midline (P_4 , P_{12} , P_{16} and P_{18}) and at the sides (P_{10} and P_{13}). Being P_{10} and P_{13} located in symmetric positions, their comparison also provides an indication of the symmetry of the loading and, to some extent, of the structural deformation as well.

Finally, a comparison between the total load measurements the in x and z direction is shown in Fig. B.29 for the tests of the 0.8 mm plate at 40 m/s, pitch angle 6° . Even in the test conditions shown in Fig. B.29, in which the horizontal velocity and the pitch angle are quite high and the thickness is low, a good repeatability is observed. Fig. B.29 also shows that t_E corresponds to the time at which F_x starts to rise and F_z takes the maximum.

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